

Hands on part

Please download the jupyter notebook from
github.com/IrisdeRuiter/radio_data_rates

Requires a python environment with matplotlib and
numpy

A RADIO ASTRONOMY TALK FOR NON- RADIO ASTRONOMERS: **PETABYTES FROM THE SKY**

Harley Wood School

Saturday July 4 2026

Iris de Ruiter



THE UNIVERSITY OF
SYDNEY



THIS TALK

Why look at the sky with a radio telescope?

Emission mechanisms

My science: radio transients

New physics from new telescopes
and novel analysis techniques

Radio telescopes

Data rates

RADIO ASTRONOMY

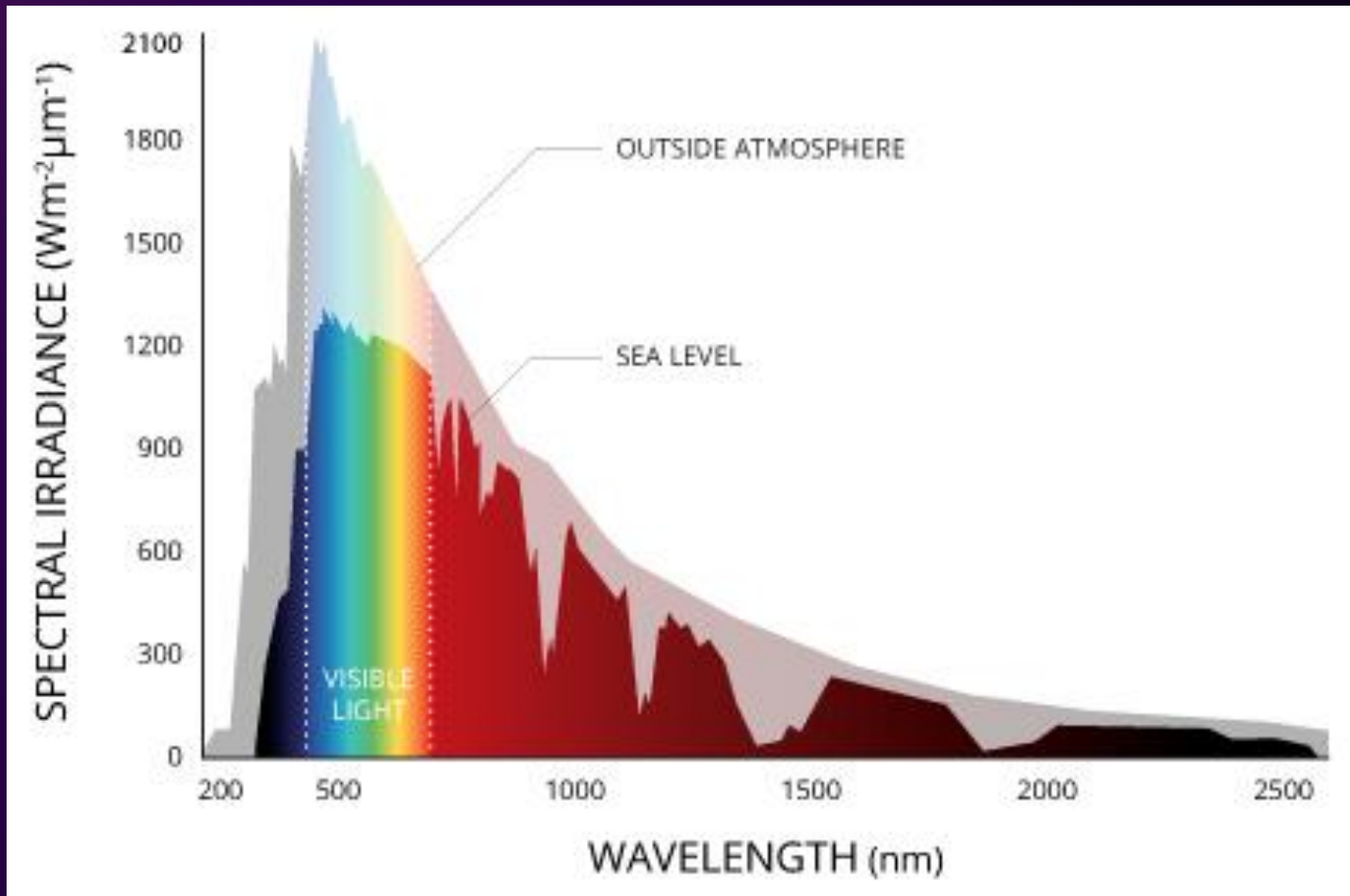
MHz and GHz frequencies / m to
cm wavelengths

Telescopes: interferometry/single
dish

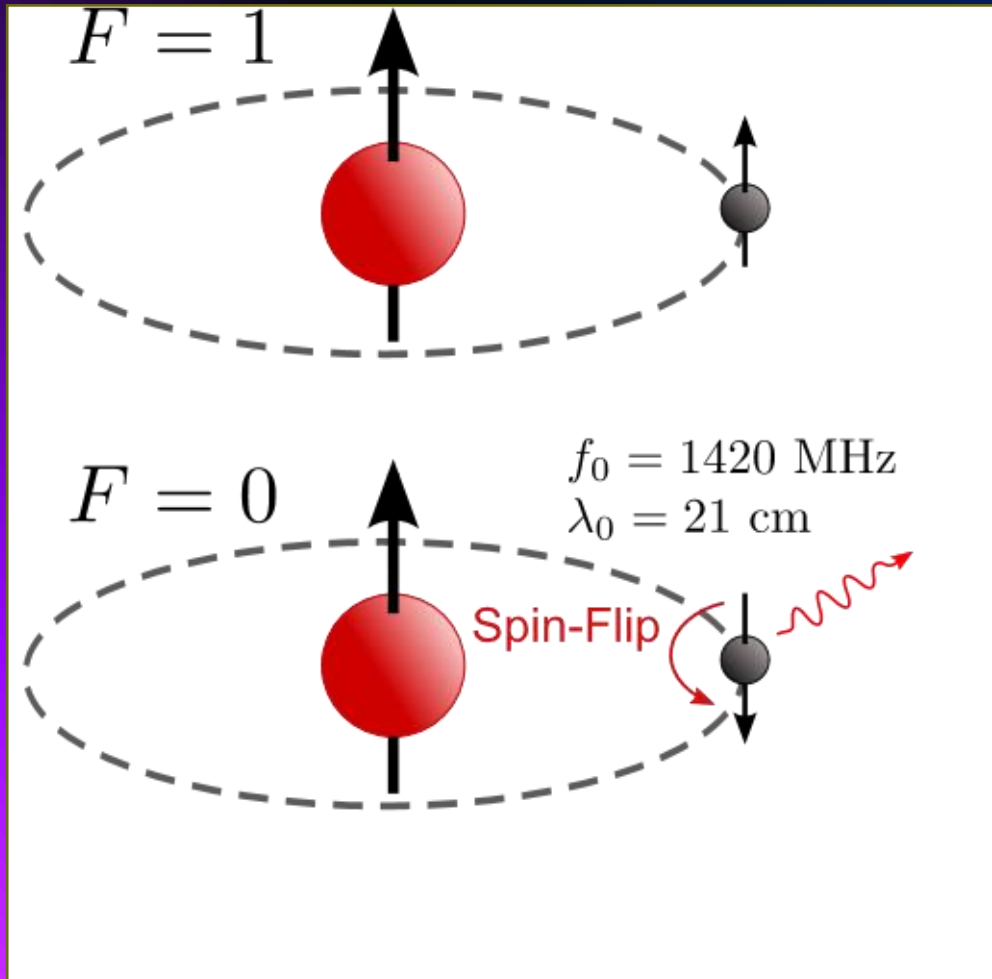
Physics: Emission mechanisms



BLACK BODY RADIATION



Typical radio
wavelength
30 cm

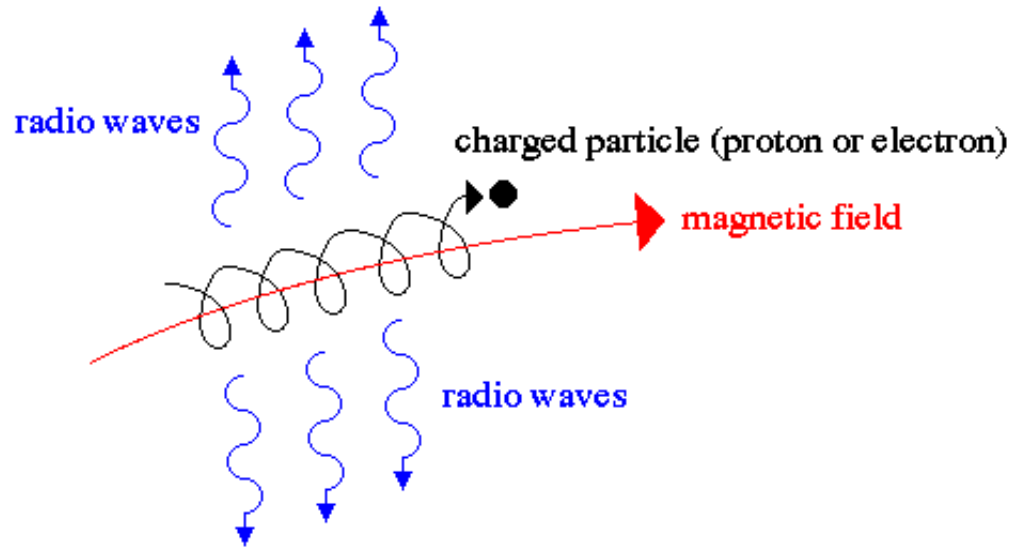


HI: THE 21 CM LINE

Hyperfine spin-flip transition
Lifetime $\sim 10^7$ years per atom

ISM contains copious amounts of
neutral hydrogen

Synchrotron radiation

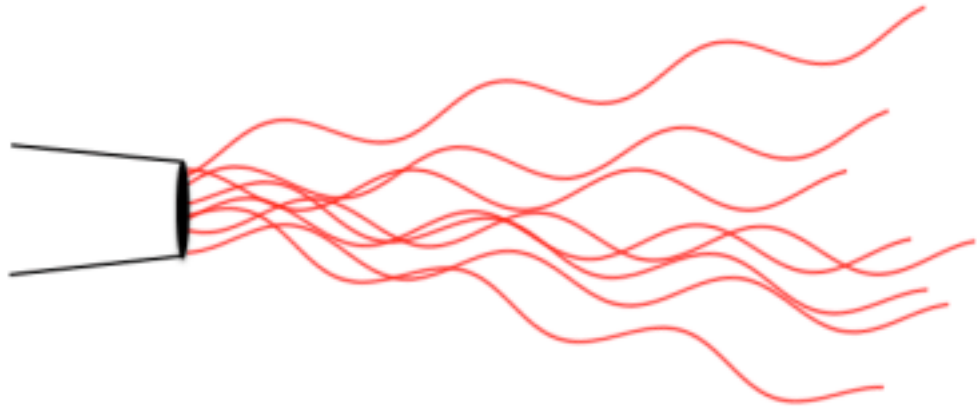


NON-THERMAL EMISSION MECHANISMS

Synchrotron emission:
Acceleration of charged particles
to relativistic speeds

Photons with the lowest energies
(ie. radio photons) – high energy
astrophysical phenomena

Energy in relativistic
particles/magnetic fields



Incoherent LED Light

(IN)COHERENT EMISSION

Incoherent emitters

$$P_{tot} = N \cdot P_{single} \sim N \cdot q^2$$

Synchrotron emission

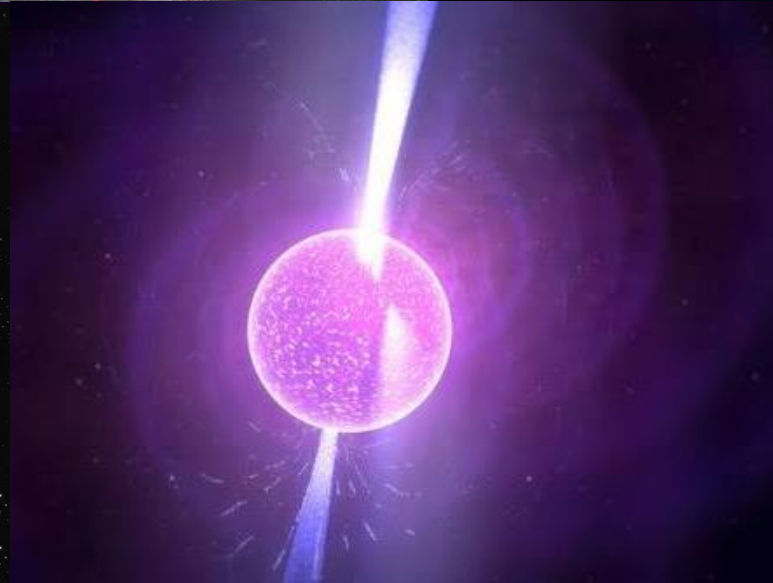
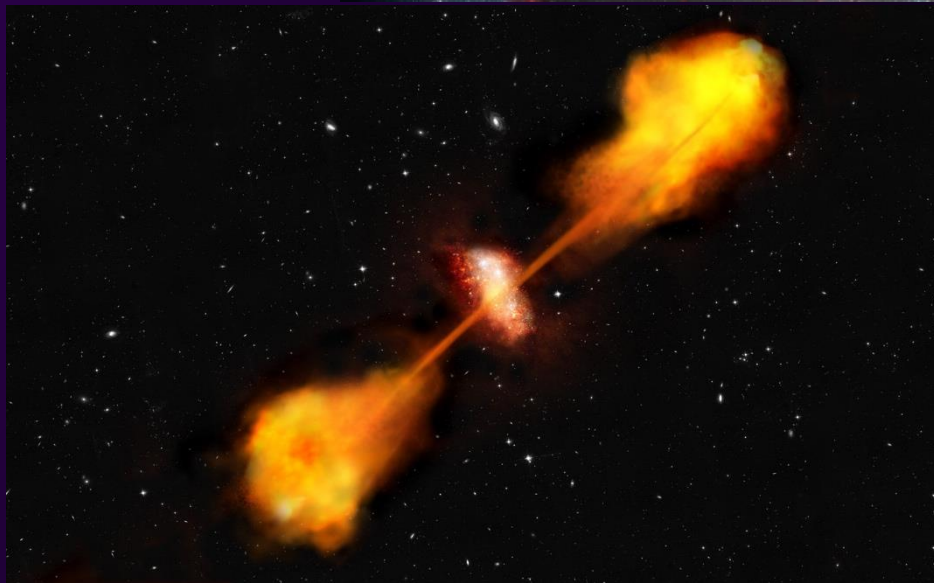
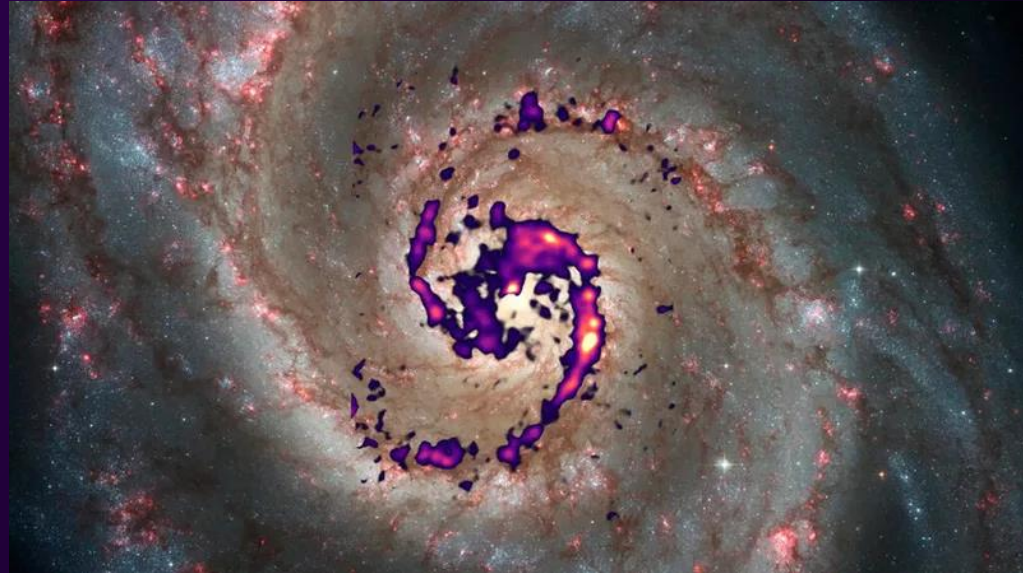
Coherent Laser Light



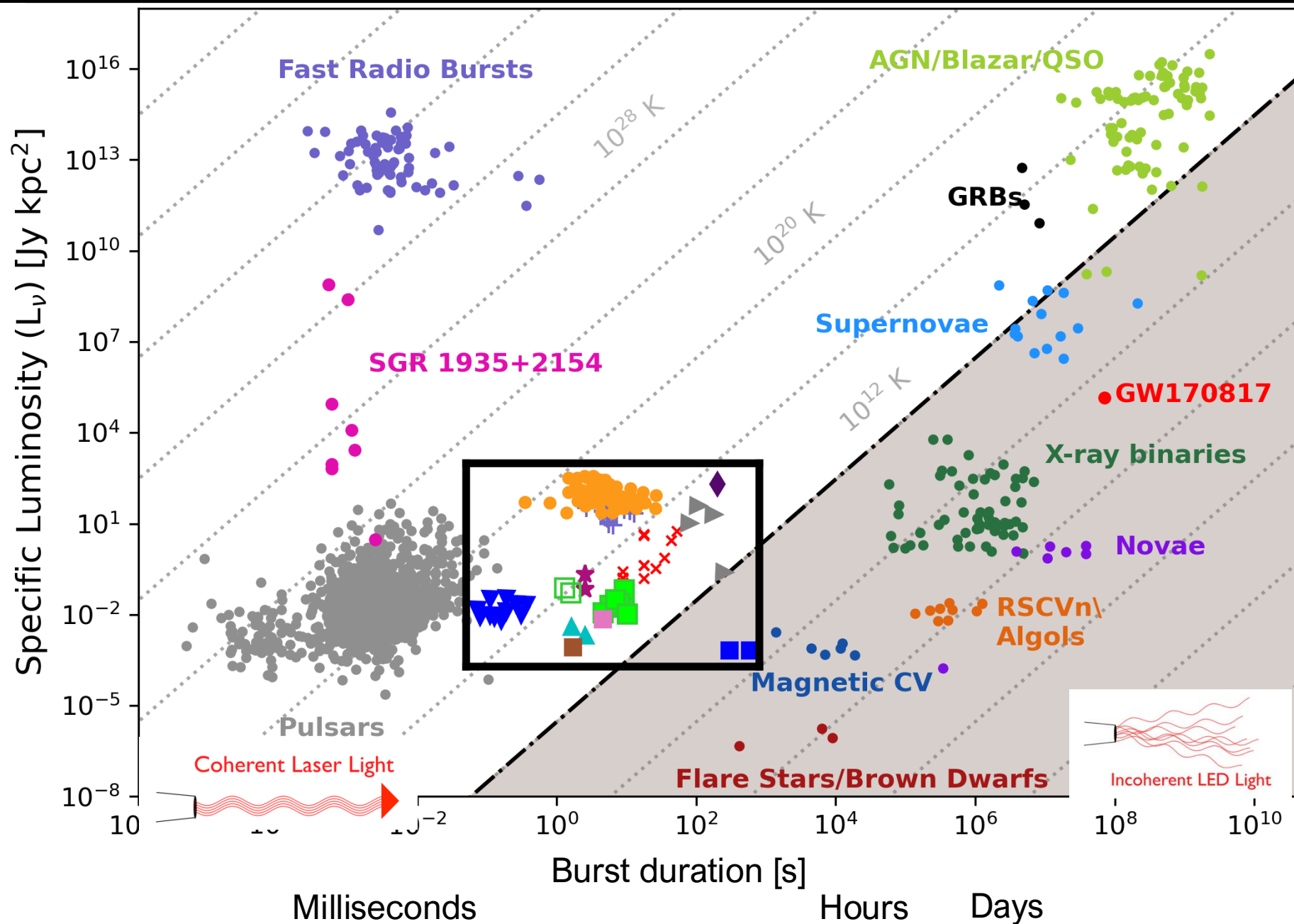
Coherent emitters: particles
'act as one'

$$P_{tot} = (N \cdot q)^2$$

Pulsars/FRBs

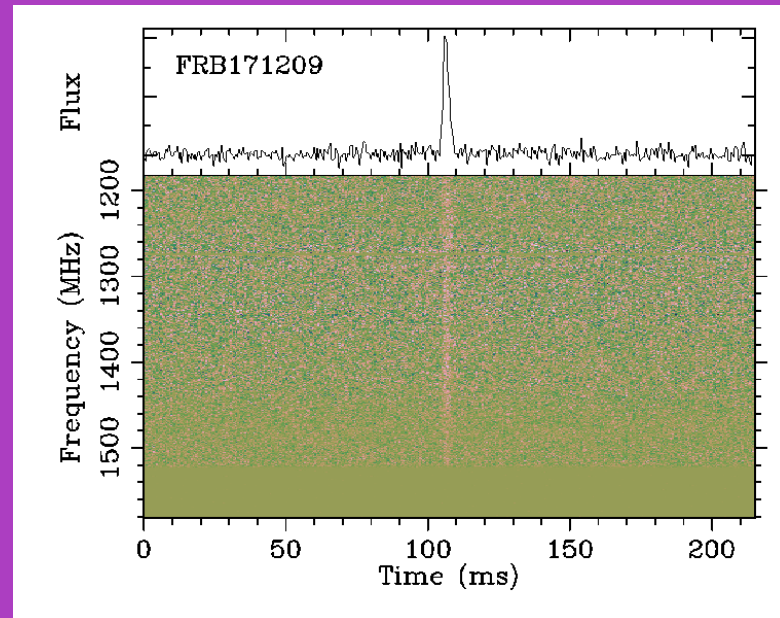


RADIO TRANSIENTS



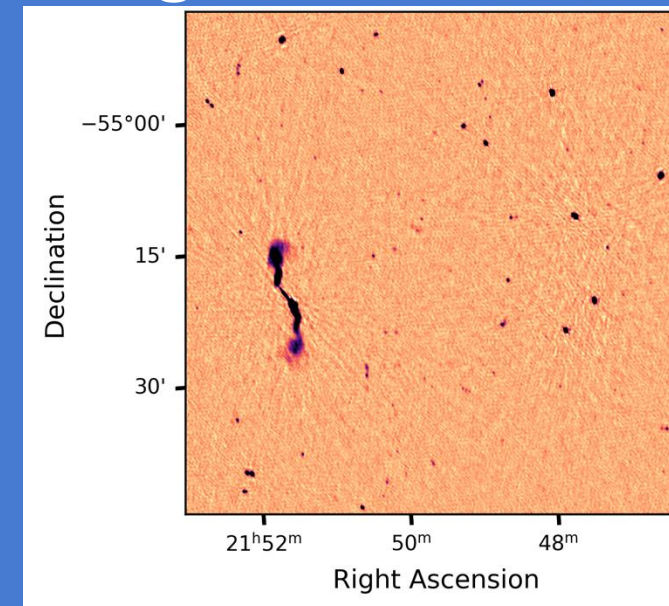
Radio transients in 2020

Beamforming
Fast
Time series

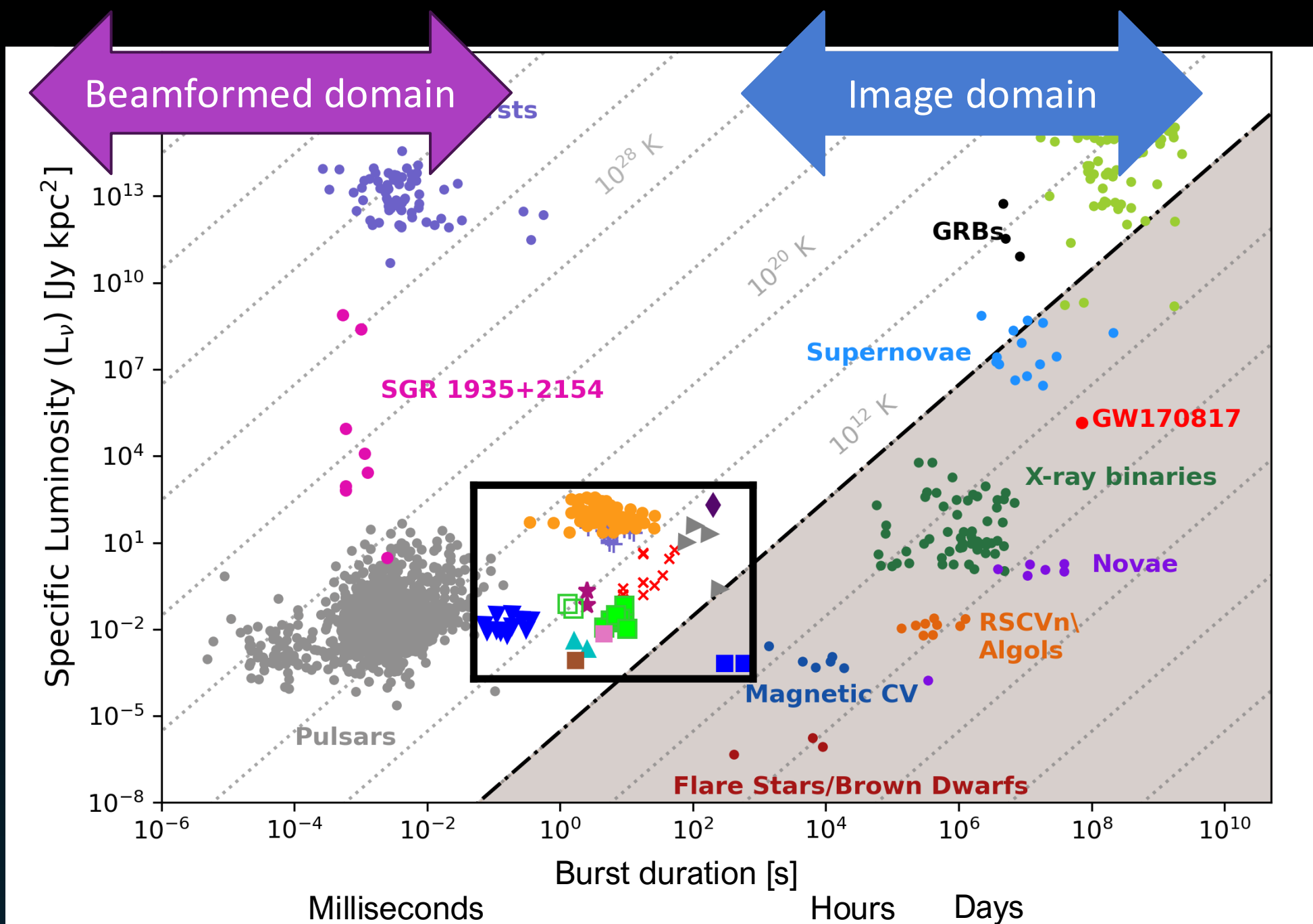


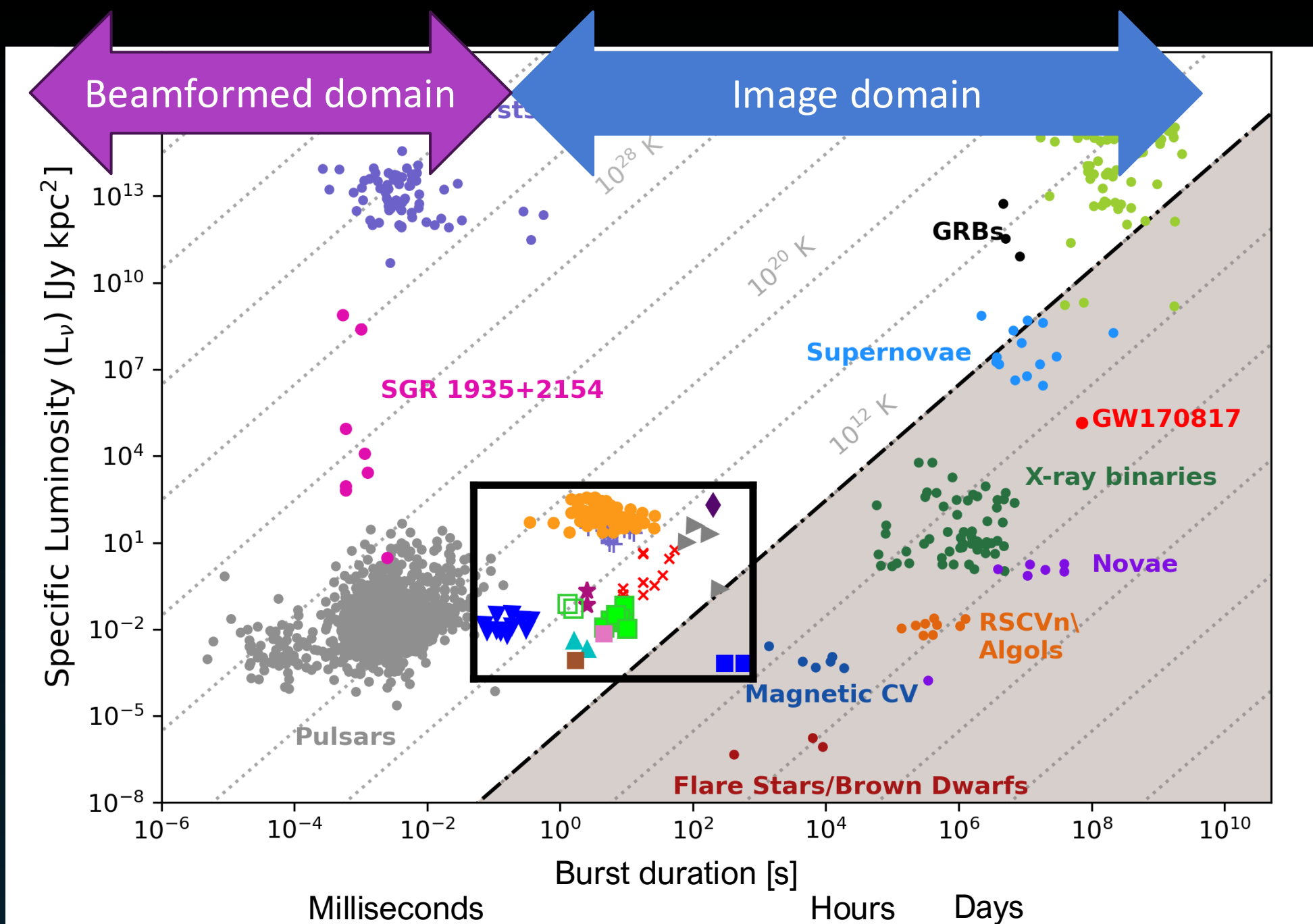
- Pulsars
- FRBs

Image domain
Slow
Images



- AGN
- Synchrotron transients





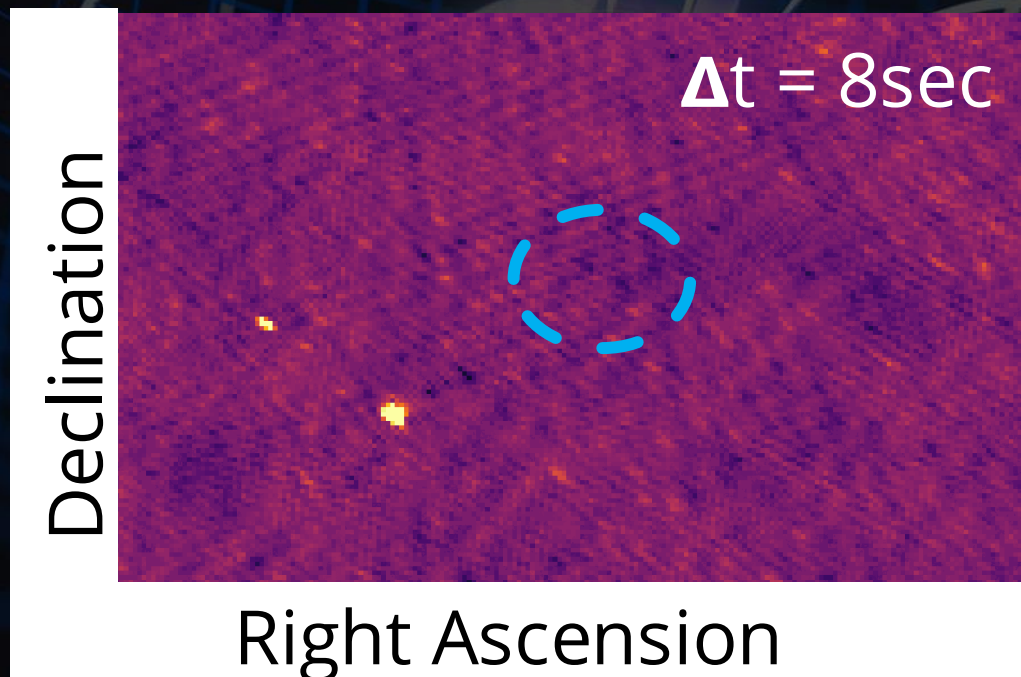
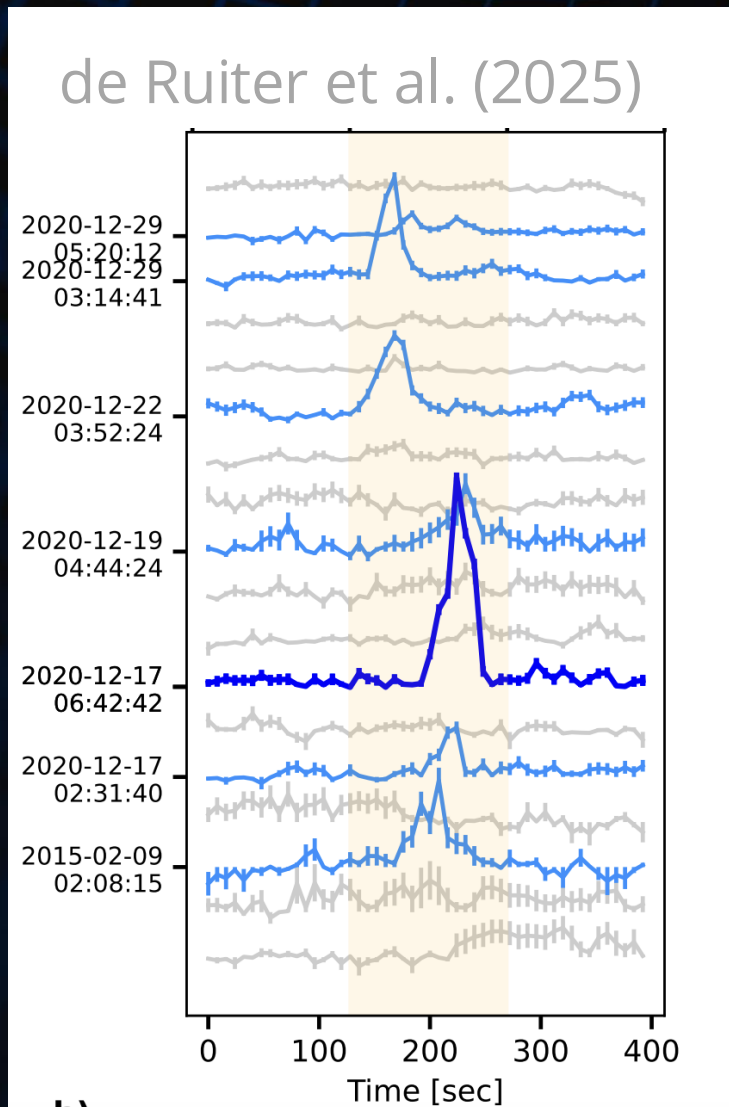
NEW TELESCOPES &
NEW ANALYSIS
METHODS
REVEAL NEW PHYSICS

Second timescale transient searches

- MeerKAT – TRON (Smirnov et al. 2025)
- MWA (Horváth et al. 2025, Hurley-Walker et al. 2026)
- ASKAP – VASTER (Y. Wang et al. 2026)
- LOFAR (de Ruiter et al. 2024)
- DSA – 110 (Sherman. et al 2025)
- CHIME/Slow (Mate et al. 2026)
- GMRT – SPOTLIGHT (Roy et al 2024)

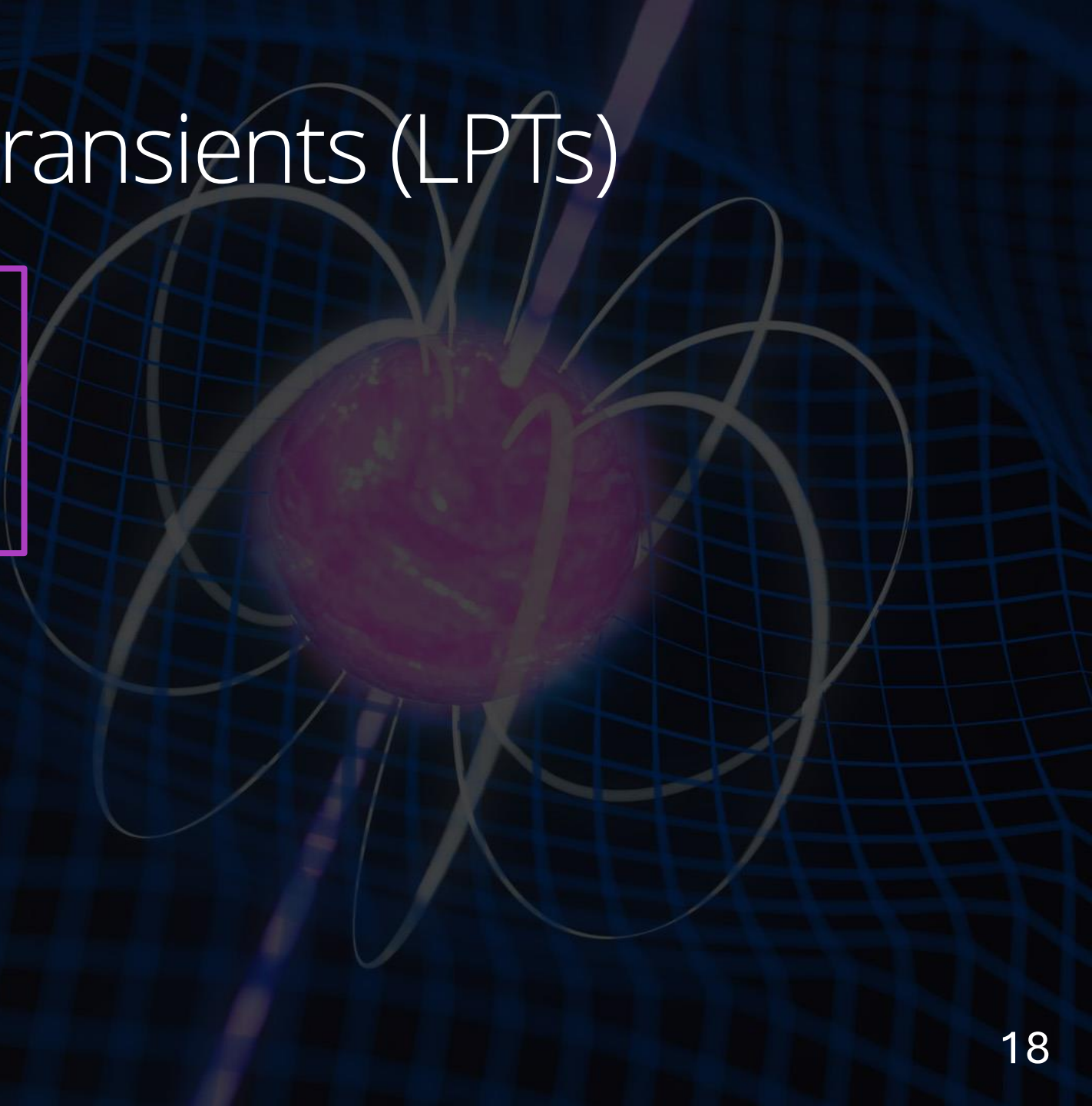
Very recent development -> new type of source!

Long-period radio transients (LPTs)



Long-period radio transients (LPTs)

- Bright second to minute radio bursts
- Highly polarised
- Periods of minutes to hours

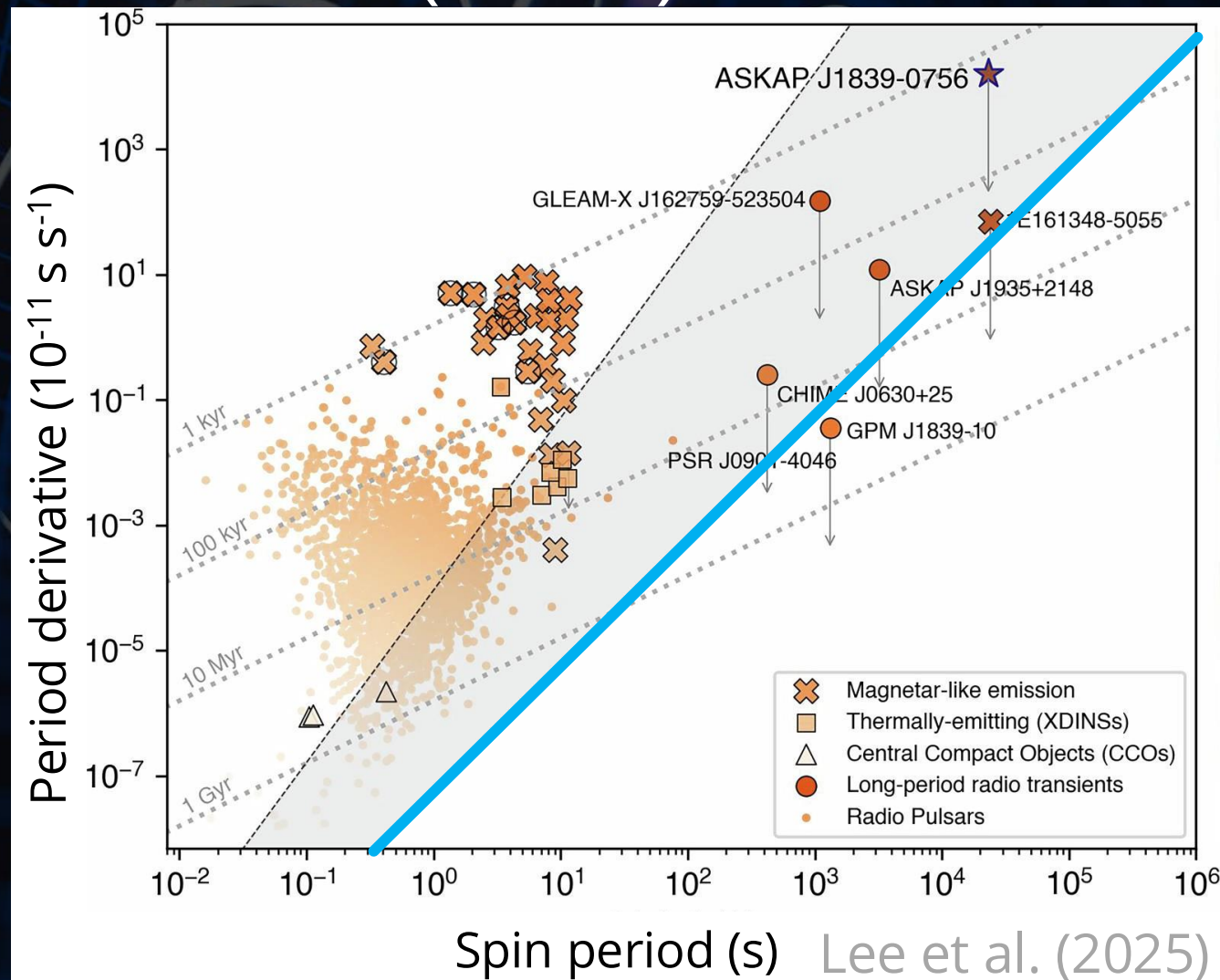


Long-period radio transients (LPTs)

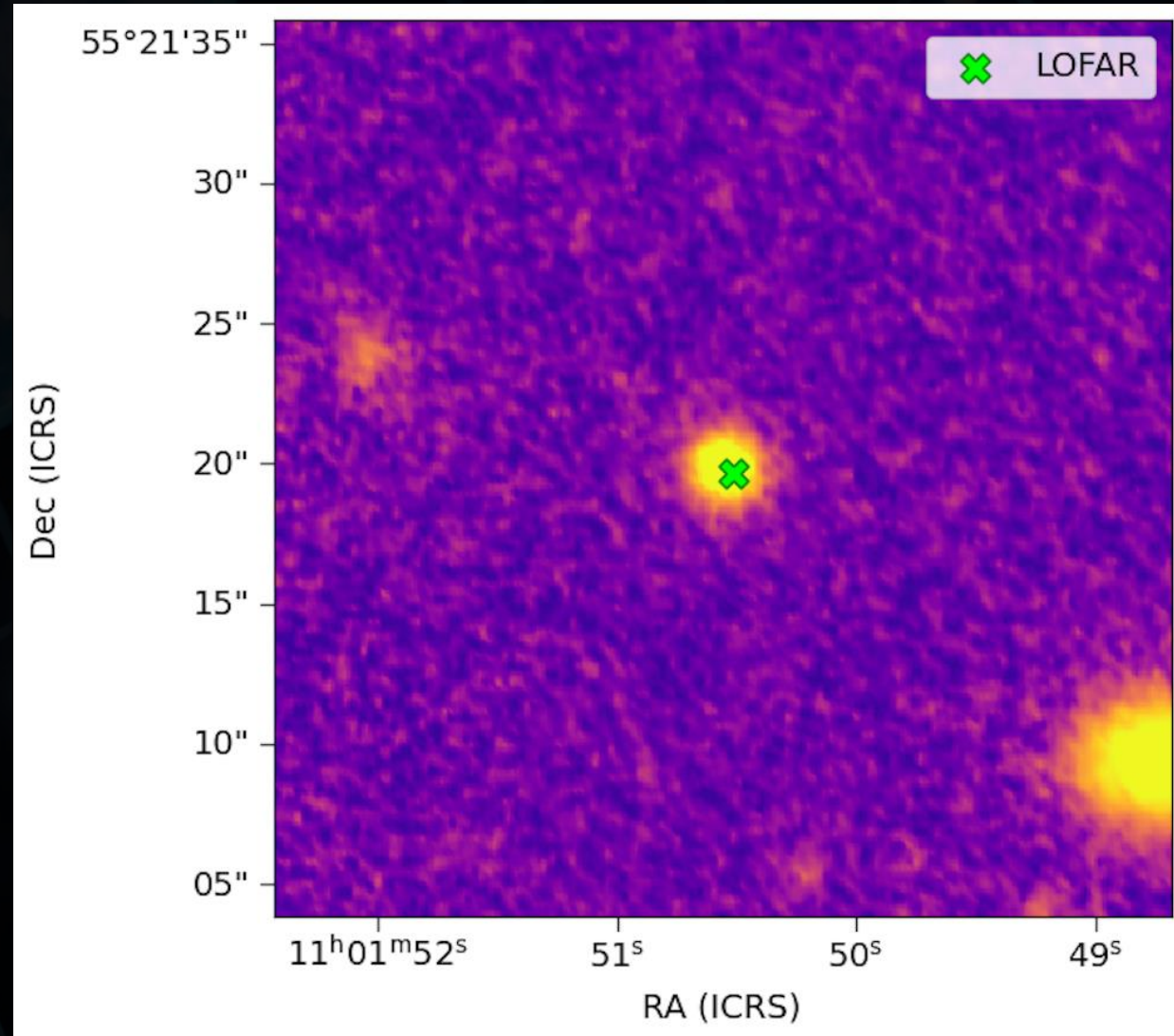
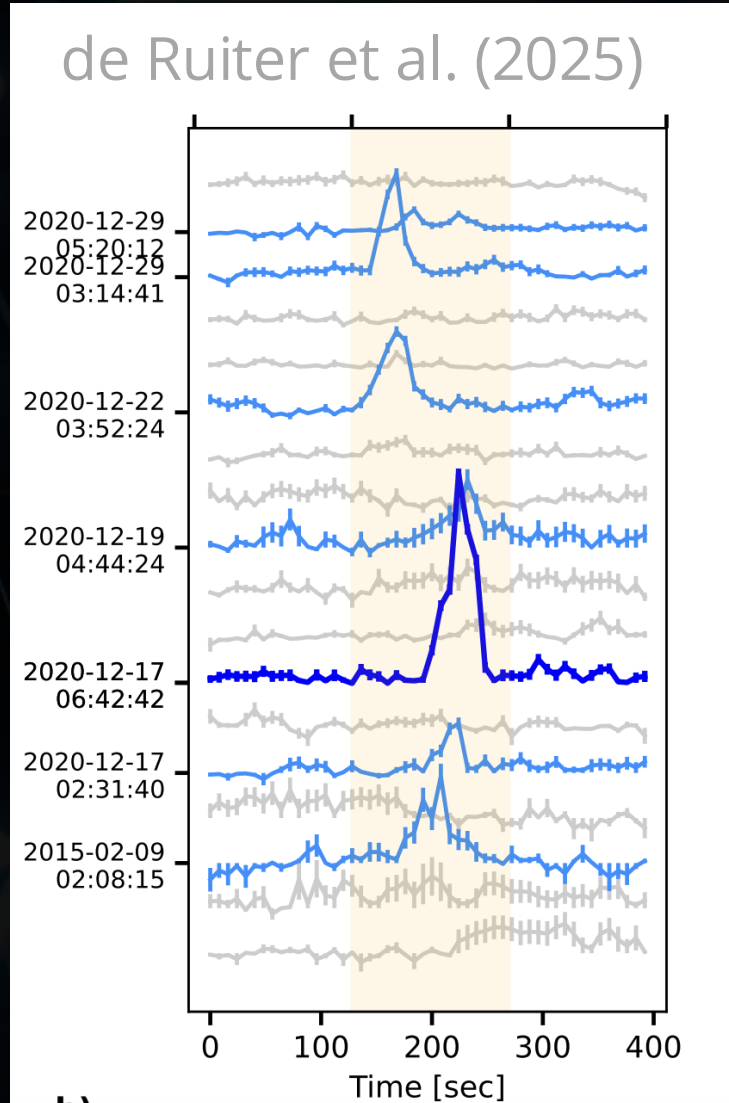
- Bright second to minute radio bursts
- Highly polarised
- Periods of minutes to hours

Slowly spinning pulsars?

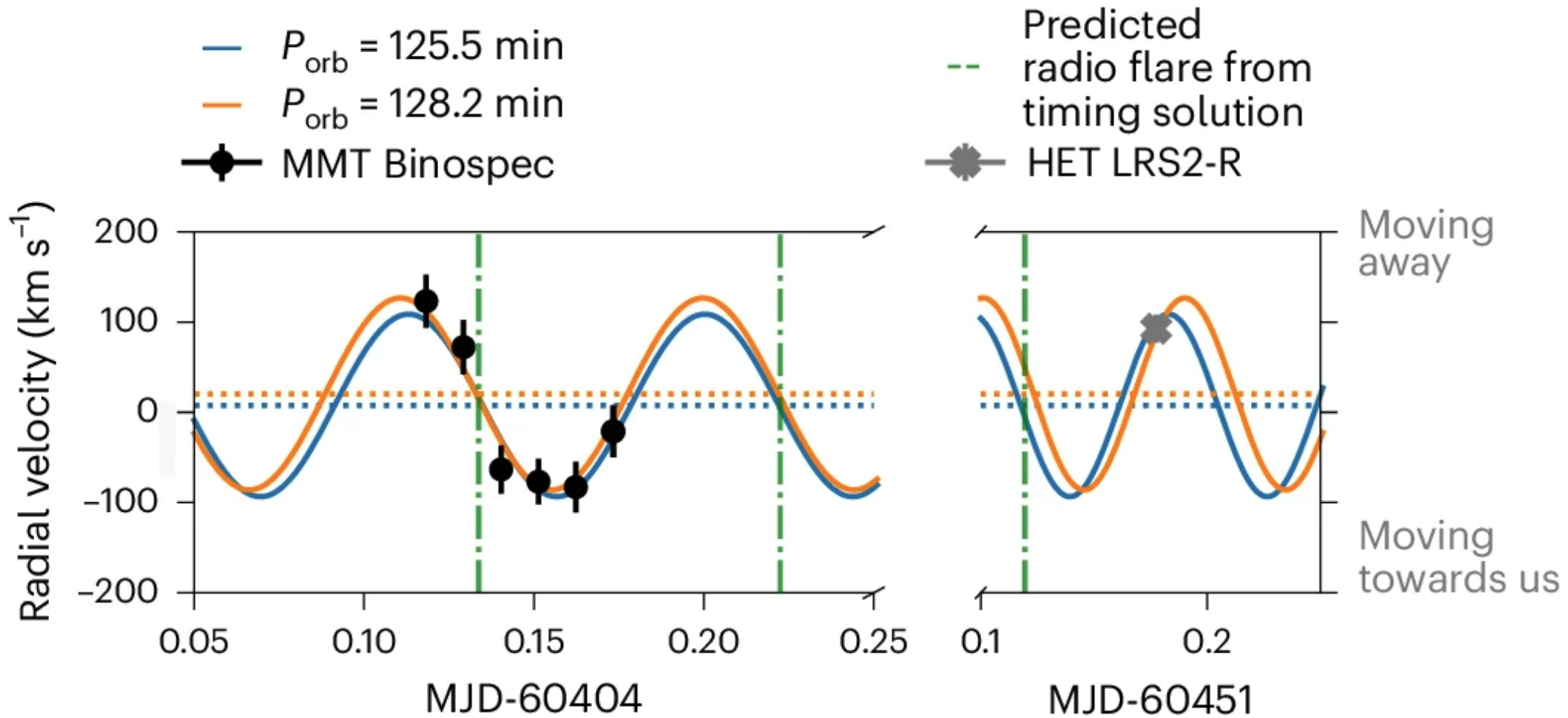
- Radio luminosity > spin-down luminosity



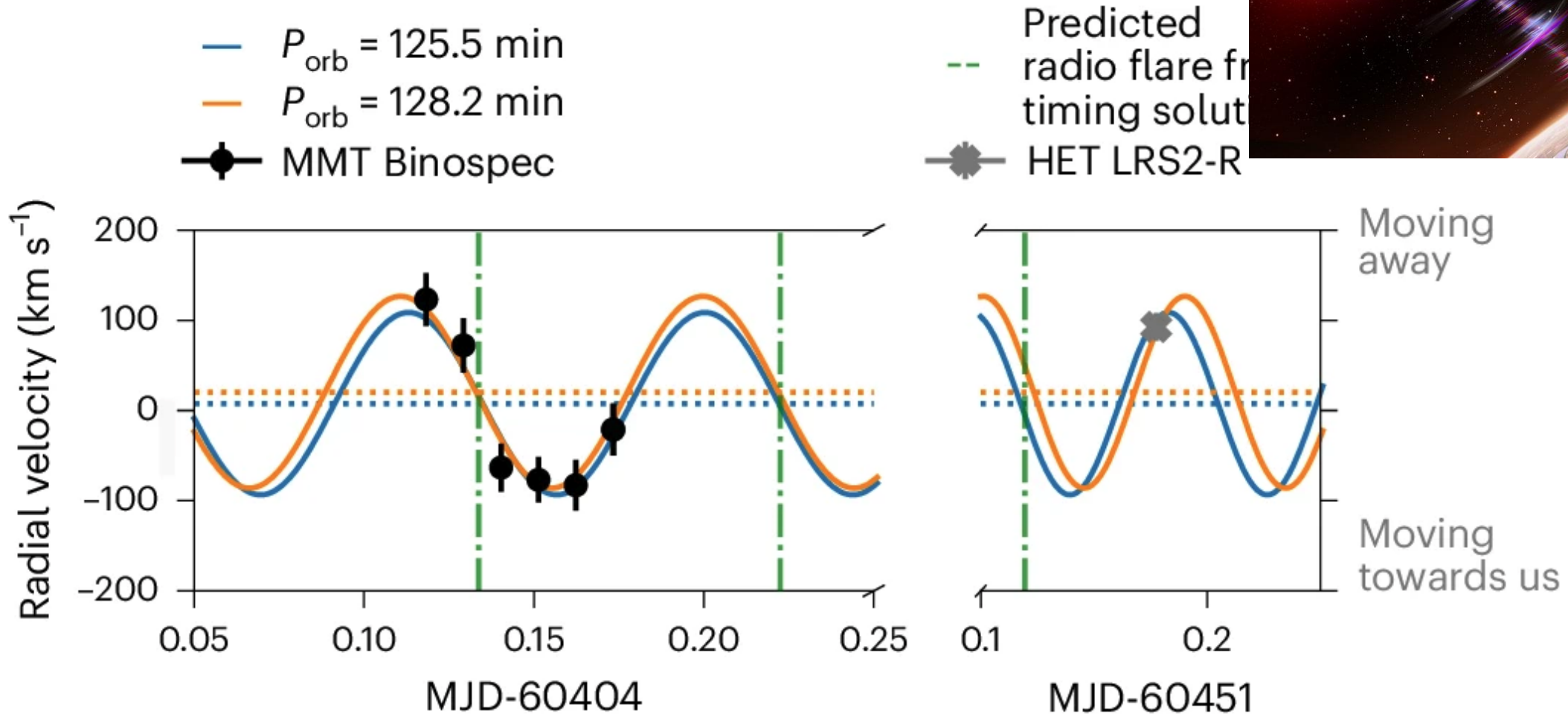
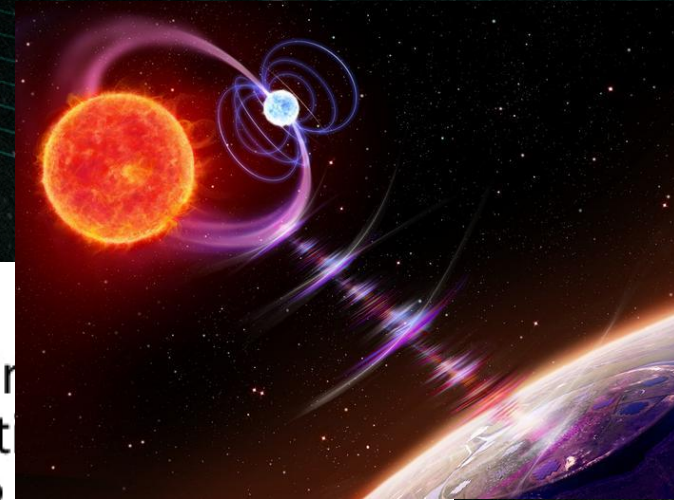
LPTs: pulsars or something else?



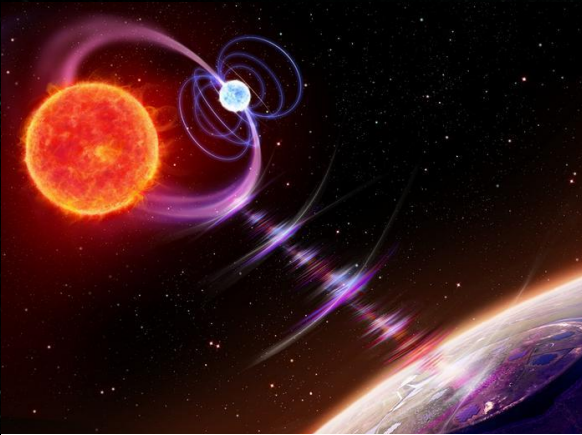
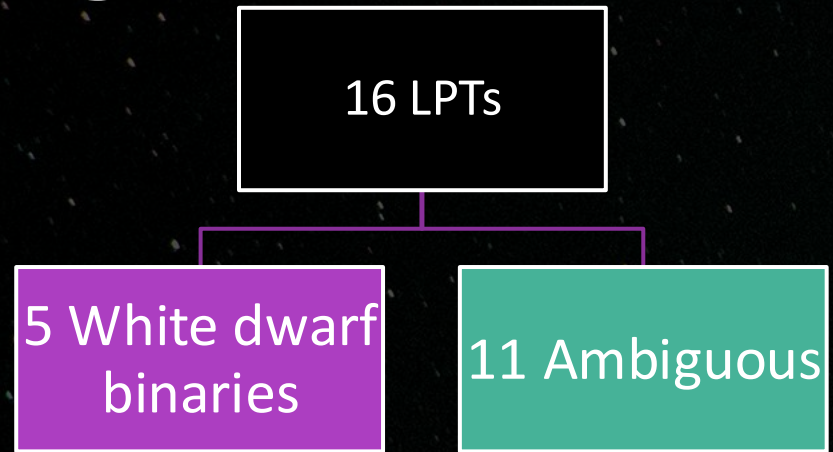
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LPTs: pulsars or something else?

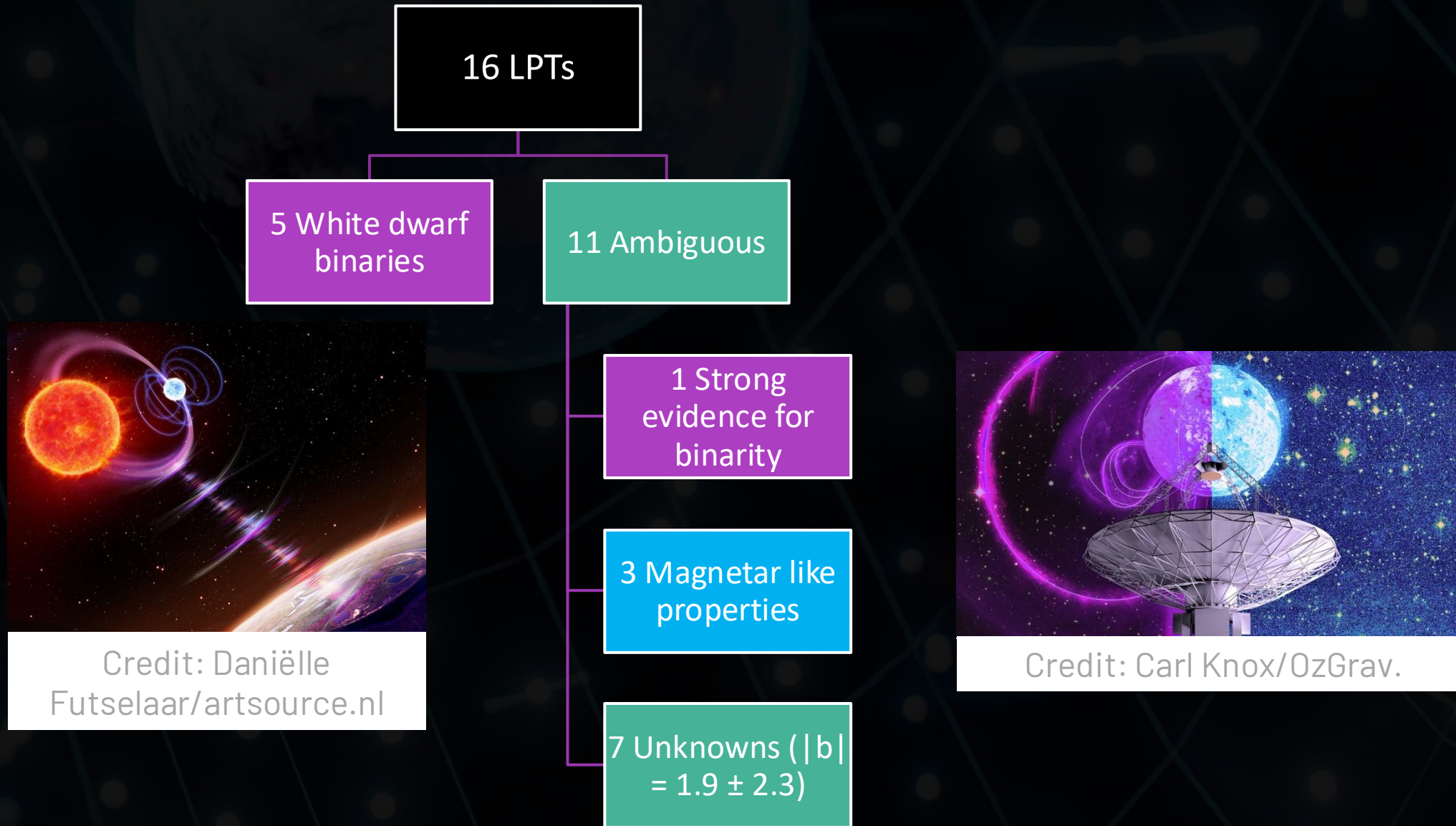


LPTs



Credit: Daniëlle
Futselaar/artsources.nl

LPTs: white dwarf binaries and pulsars?



NEW PHYSICS

1. WHITE DWARFS CAN ACT LIKE PULSARS (WHEN IN A BINARY)
2. HOW DO WE GET SLOWLY SPINNING PULSARS?

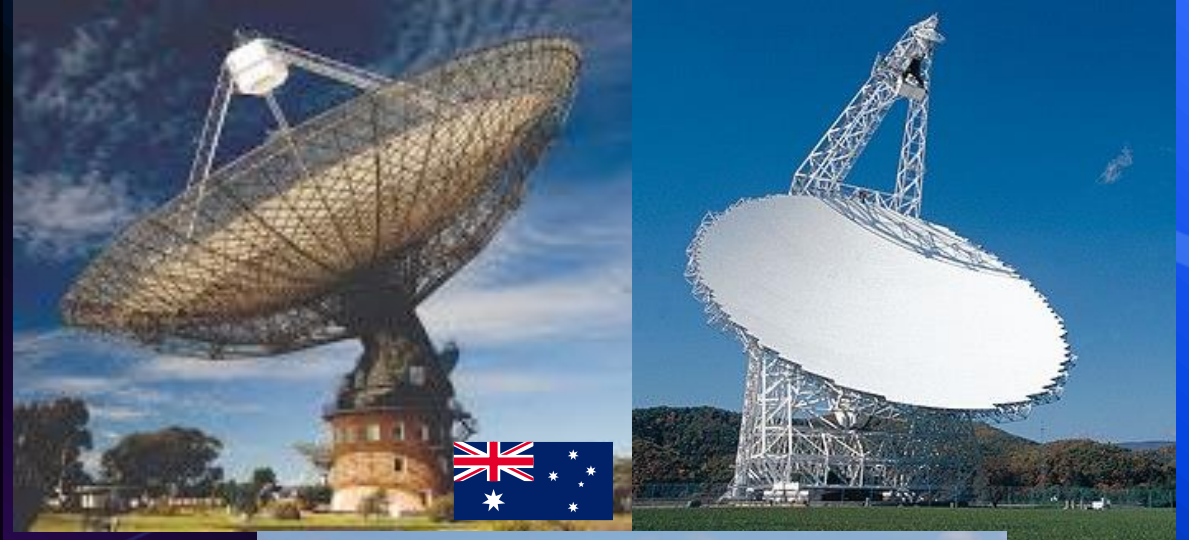
RADIO TELESCOPES
PRODUCE BIG DATA

RADIO TELESCOPES – SINGLE DISH

Murriyang in Parkes - 64 m

Effelsberg, Greenbank – 100 m

Sensitivity $\sim D^2$

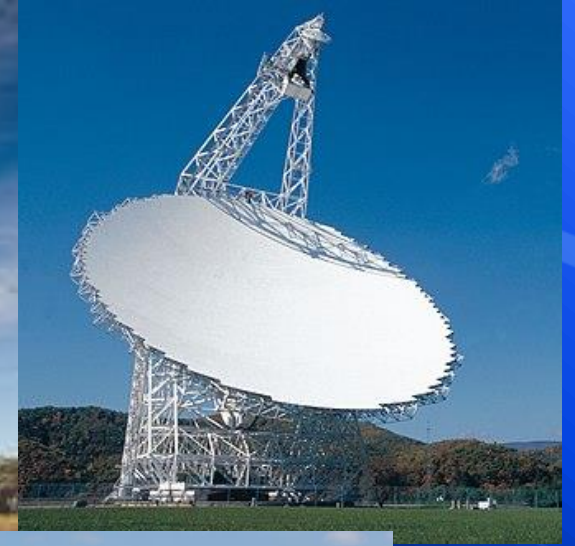
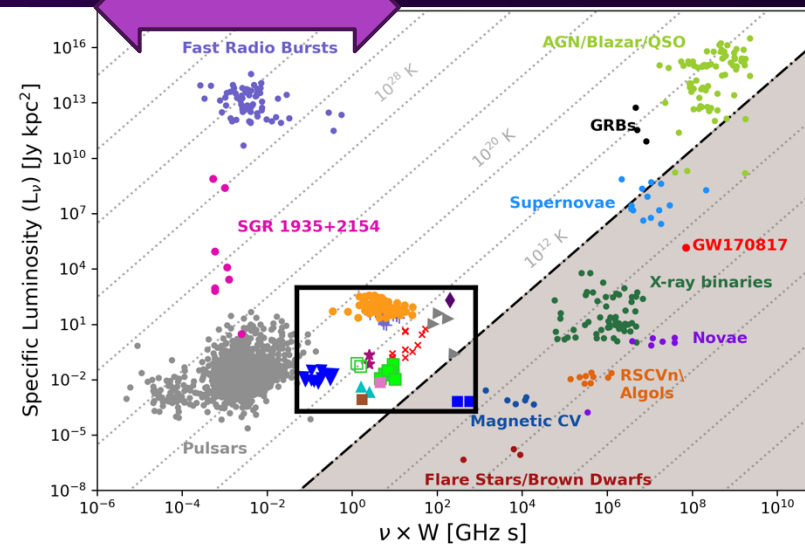


RADIO TELESCOPES – SINGLE DISH

Angular resolution limit

$$\theta \sim \frac{\lambda}{D}$$

$$\theta \sim \frac{0.3}{100} = 0.2 \text{ rad} = 10 \text{ amin}$$

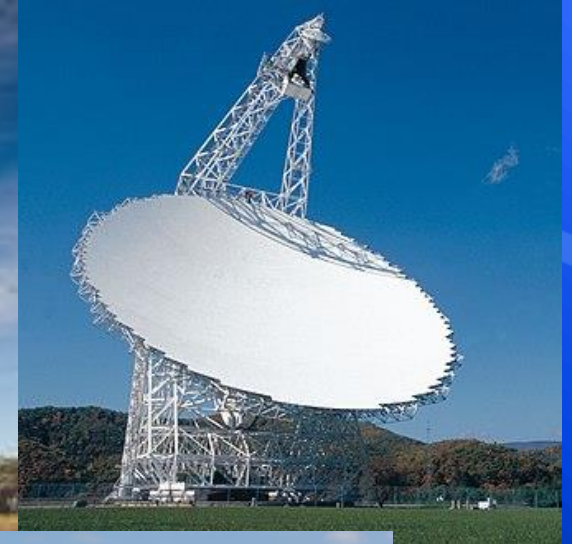


RADIO TELESCOPES – SINGLE DISH

Angular resolution limit

$$\theta \sim \frac{\lambda}{D} \quad \theta \sim \frac{0.3}{100} = 0.2 \text{ rad} = 10 \text{ amin}$$

FAST – 500 meter



RADIO TELESCOPES – INTERFEROMETRY

Typical optical image, sub
arcsecond resolution.

For $\theta = 1 \text{ asec}$

$$D \sim \frac{\lambda}{\theta} \sim \frac{0.3}{4.8 \cdot 10^{-6}} = 62 \text{ km}$$

RADIO TELESCOPES – INTERFEROMETRY

Typical optical image, sub
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For $\theta = 1 \text{ asec}$

$$D \sim \frac{\lambda}{\theta} \sim \frac{0.3}{4.8 \cdot 10^{-6}} = 62 \text{ km}$$

Increased resolution comes
at a cost -> enormous data rates!



NEXT GENERATION TELESCOPES

Square Kilometre Array:

- SKA-low: 130.000 antennas in Murchison shire in WA
- SKA-mid: 200 steerable dishes Karoo region South Africa

DSA-2000

ngVLA



RADIO TELESCOPES PRODUCE BIG DATA

Hands on:

github.com/IrisdeRuiter/radio_data_rates

SUMMARY

1. Radio astronomy offers a unique perspective
 1. neutral hydrogen
 2. magnetic fields and charged particles
2. New telescopes and new analysis techniques lead to new physics
 1. white dwarfs can act like pulsars?
3. Radio interferometers are big data machines

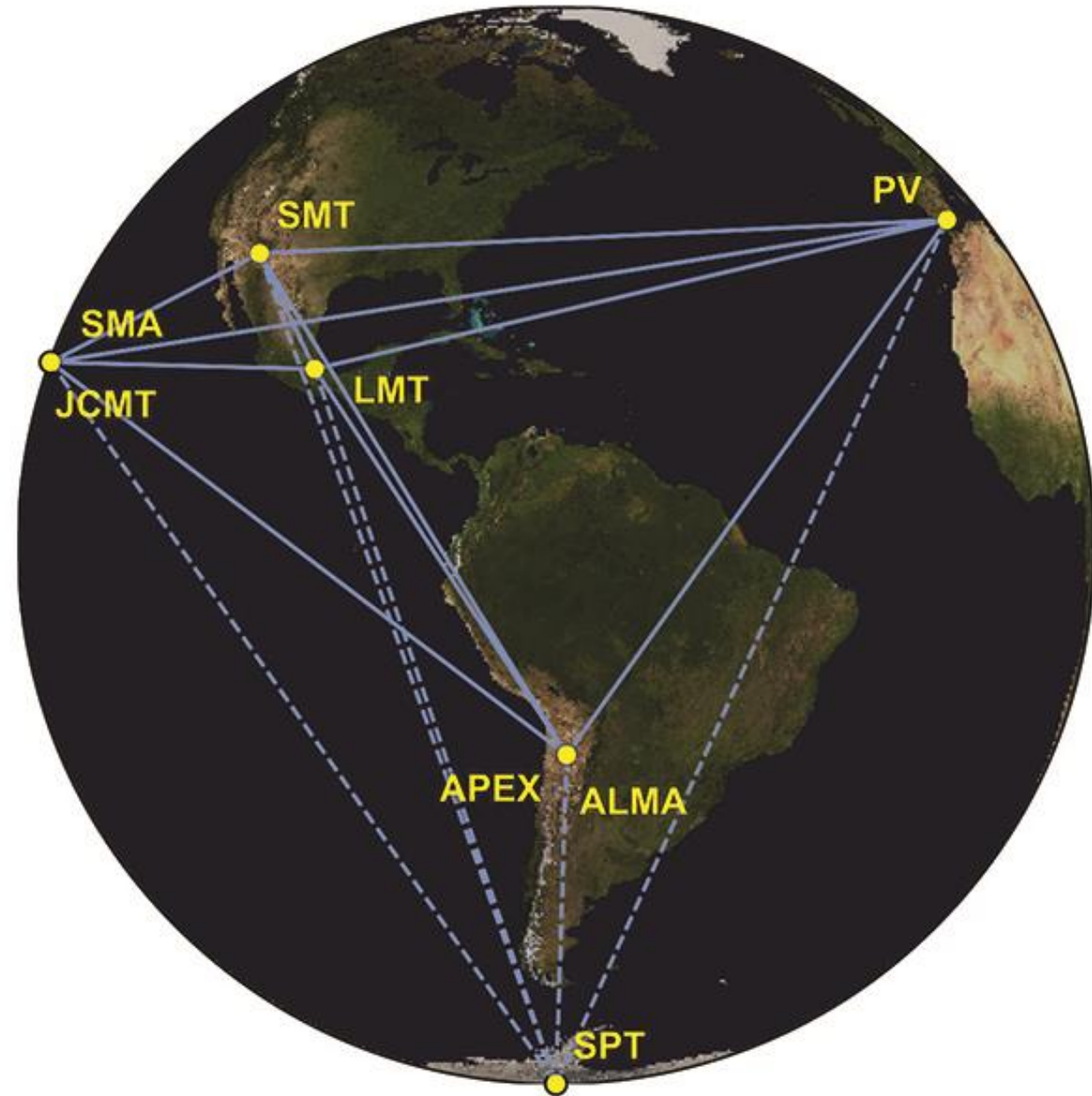
RADIO TELESCOPES – INTERFEROMETRY

EHT

$D = 10\,000\text{ km}$

$\lambda = 1.3\text{ mm}$

$$\theta \sim \frac{\lambda}{D} \sim \frac{1.3 \cdot 10^{-3}}{10 \cdot 10^6} = 25\ \mu\text{as}$$



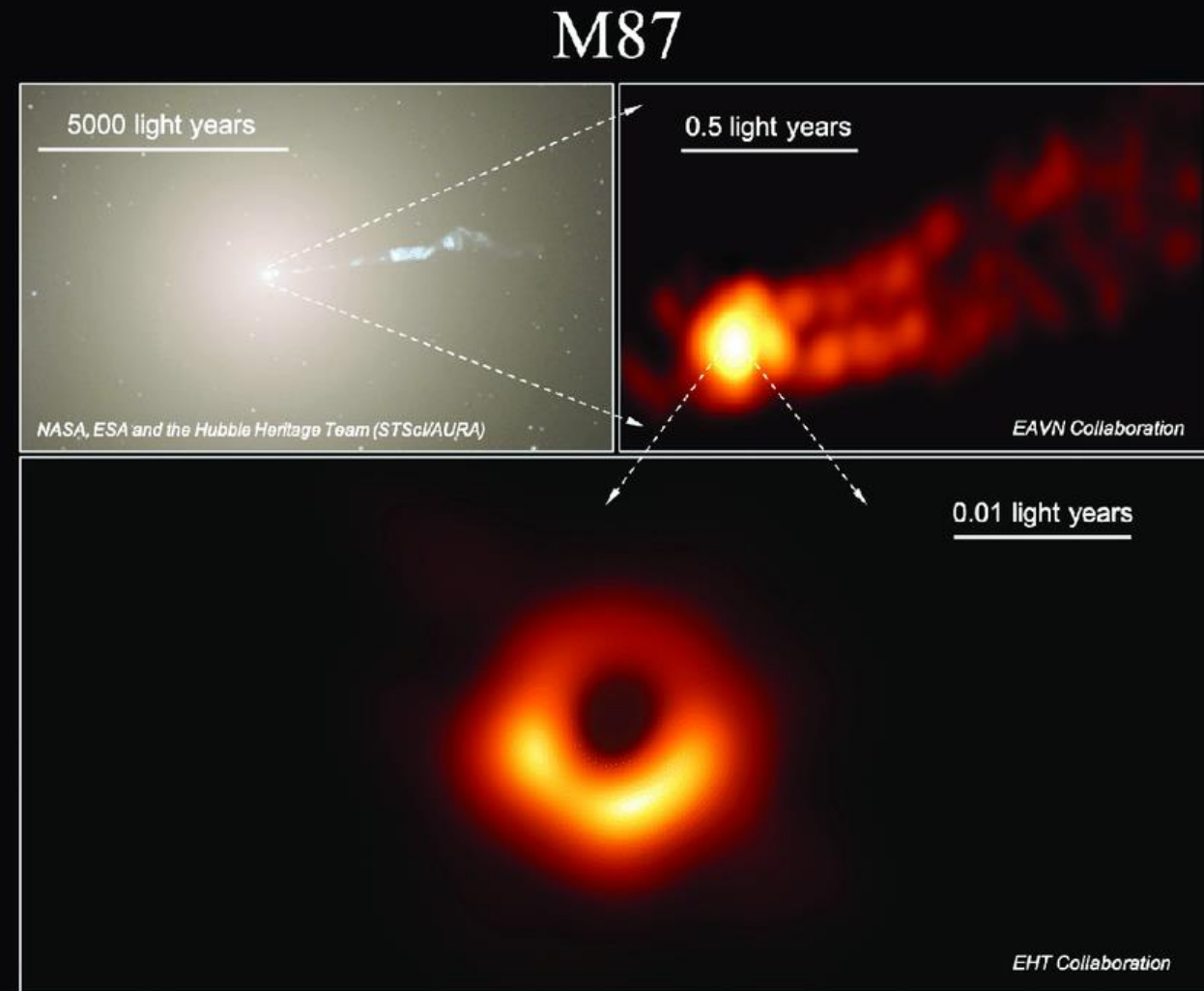
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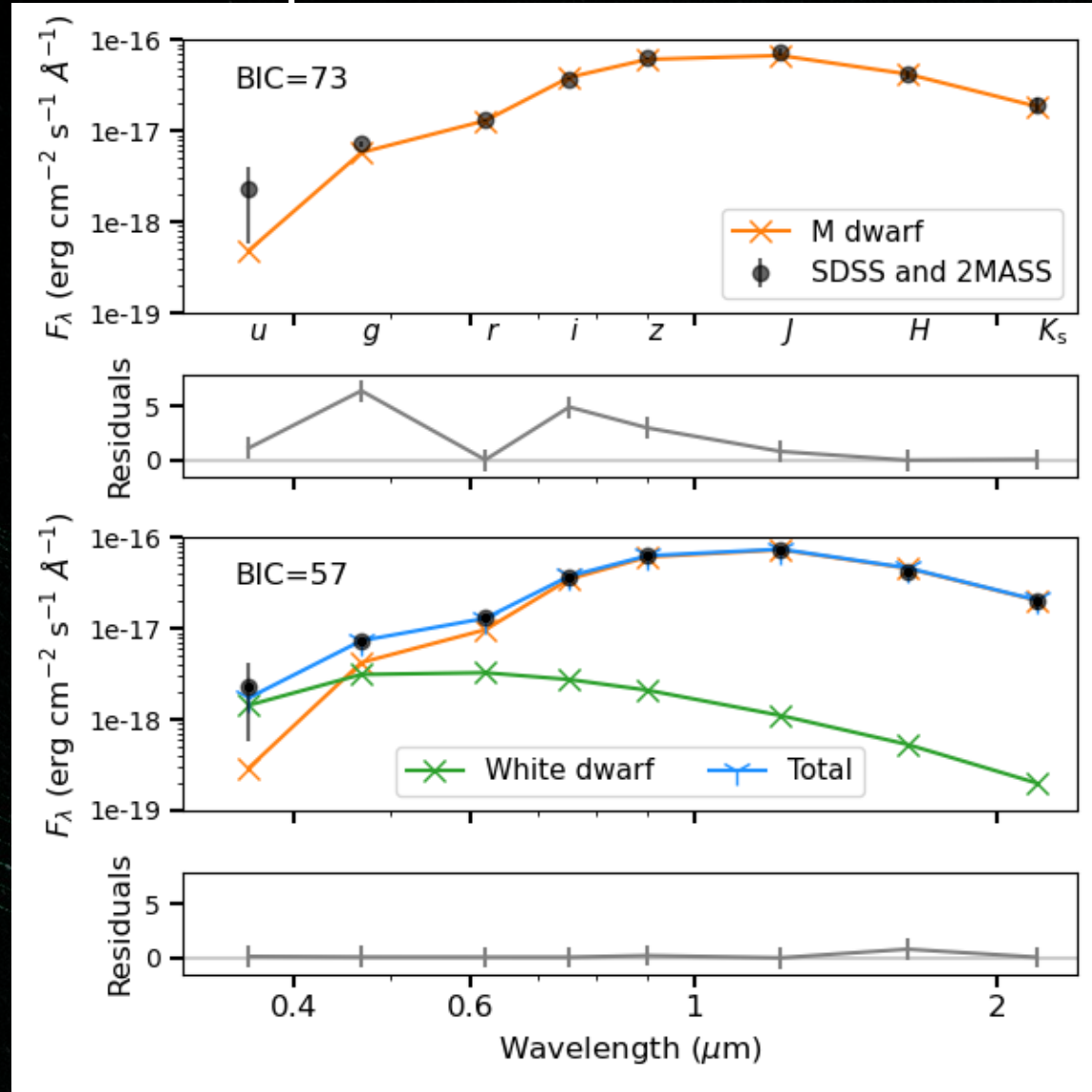
$$D \sim \frac{\lambda}{\theta} \sim \frac{1.3 \cdot 10^{-3}}{10 \cdot 10^{-6}} = 25\text{ }\mu\text{as}$$



YET ANOTHER UNIT

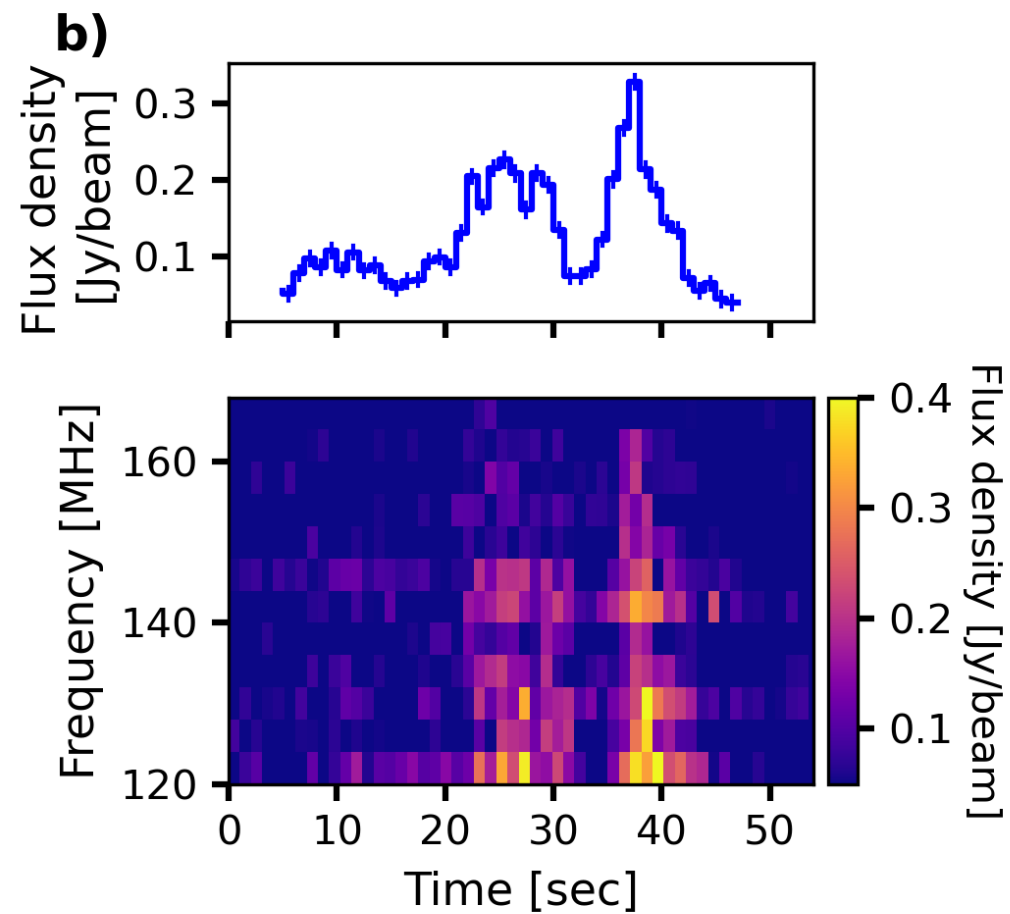
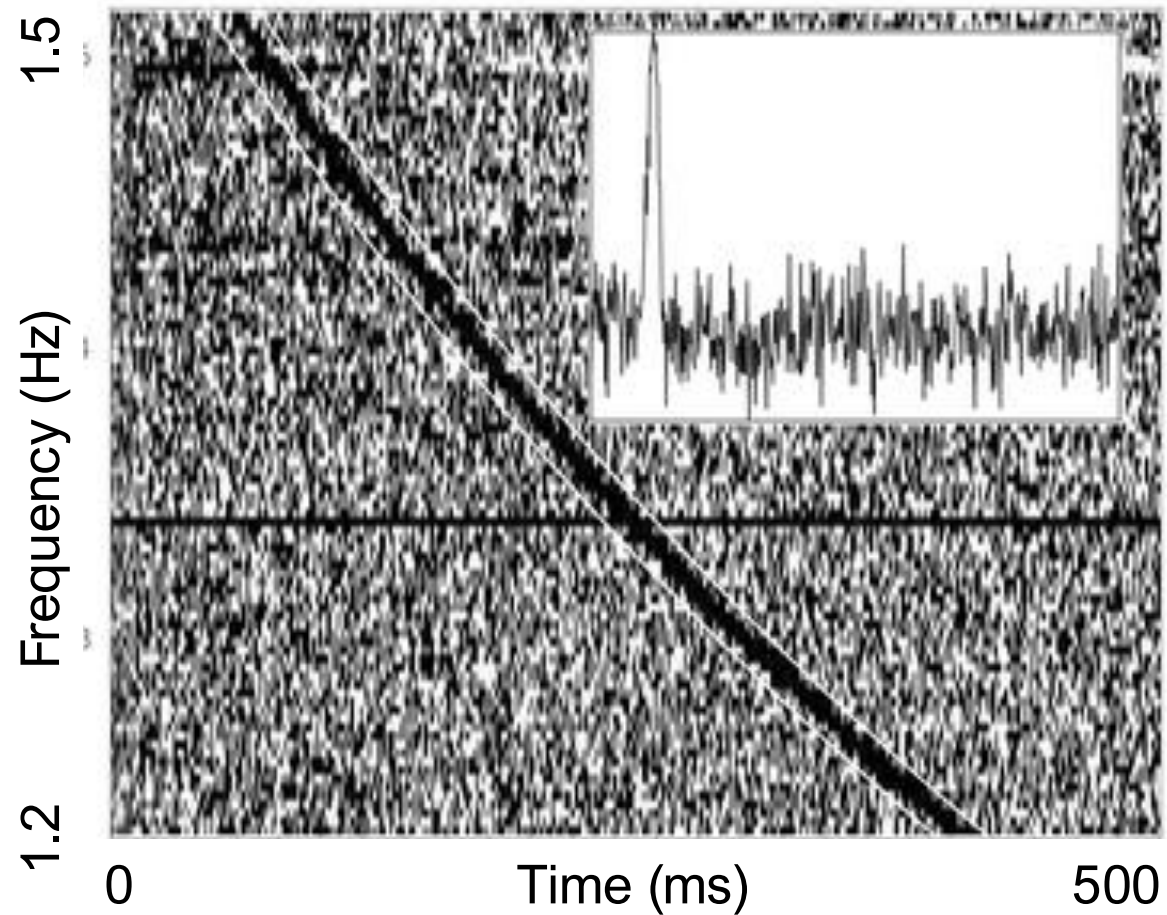
- Spectral flux density, or spectral irradiance
- Jansky = 1 Jy =
- $10^{-23} \text{ erg} \cdot \text{s}^{-1} \cdot \text{cm}^{-2} \cdot \text{Hz}^{-1}$
- $10^{-26} \text{ W} \cdot \text{m}^{-2} \cdot \text{Hz}^{-1}$

LPTs: pulsars or something else?

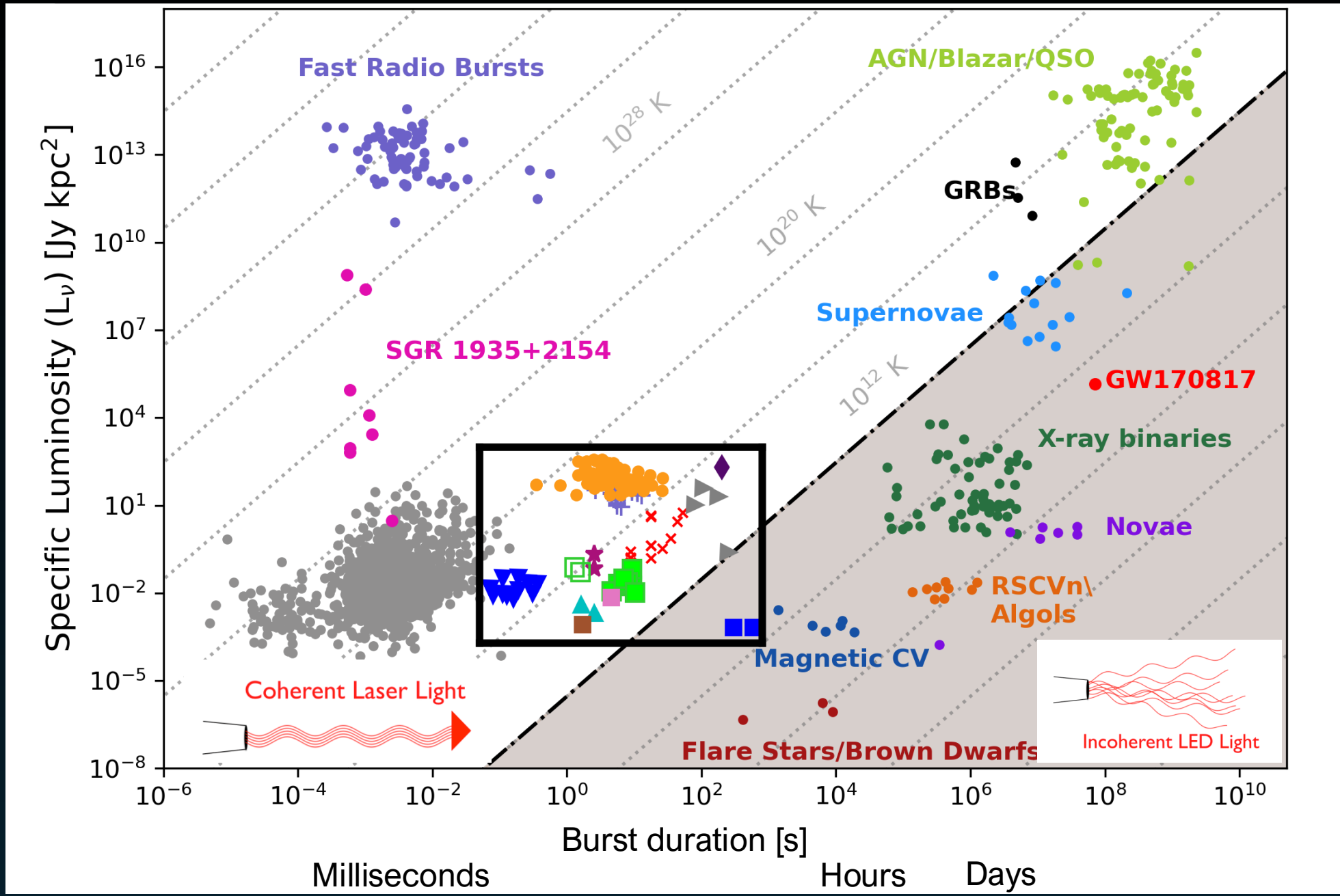


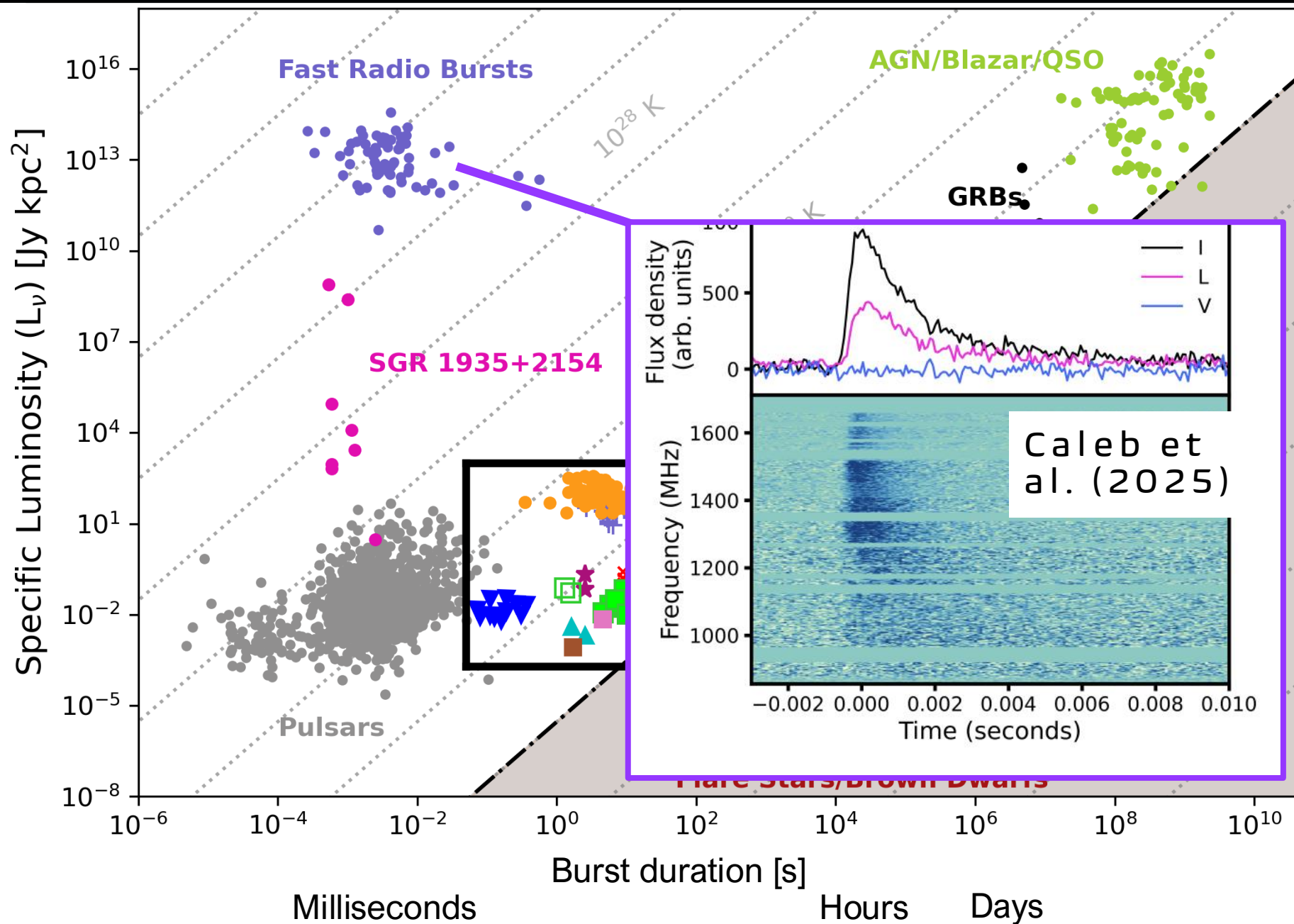
DISPERSION

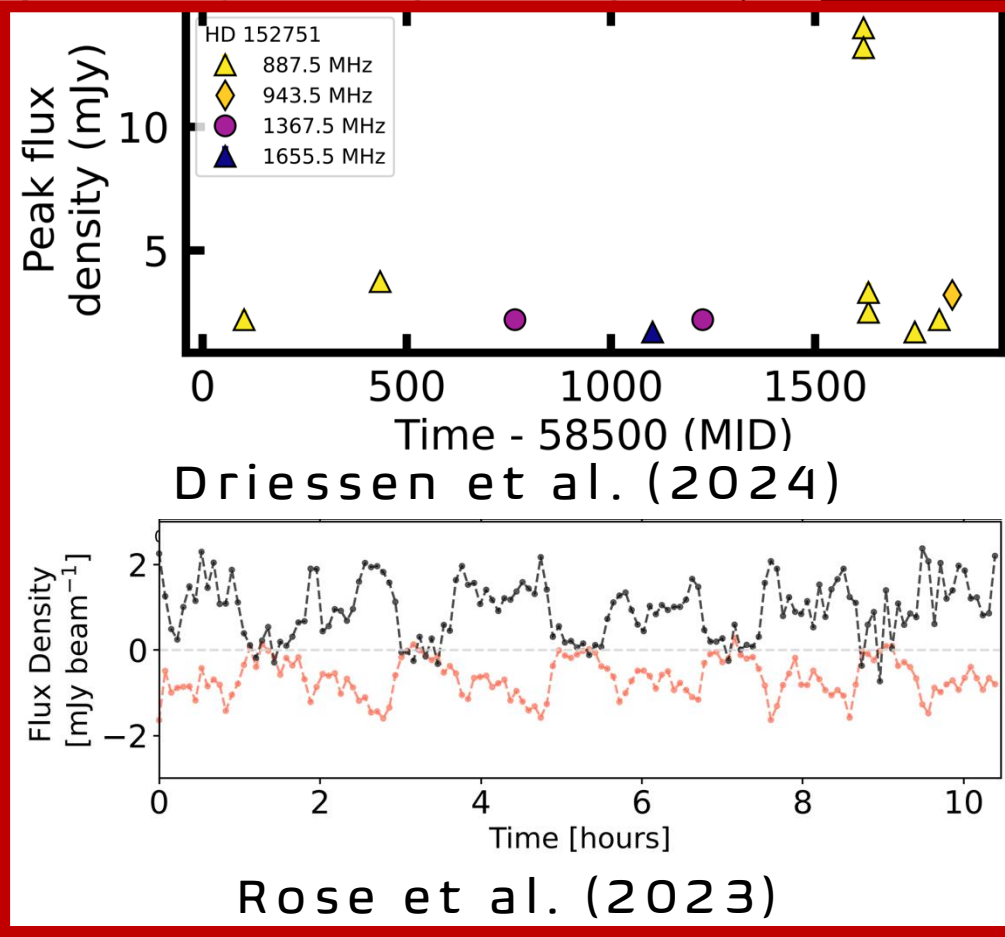
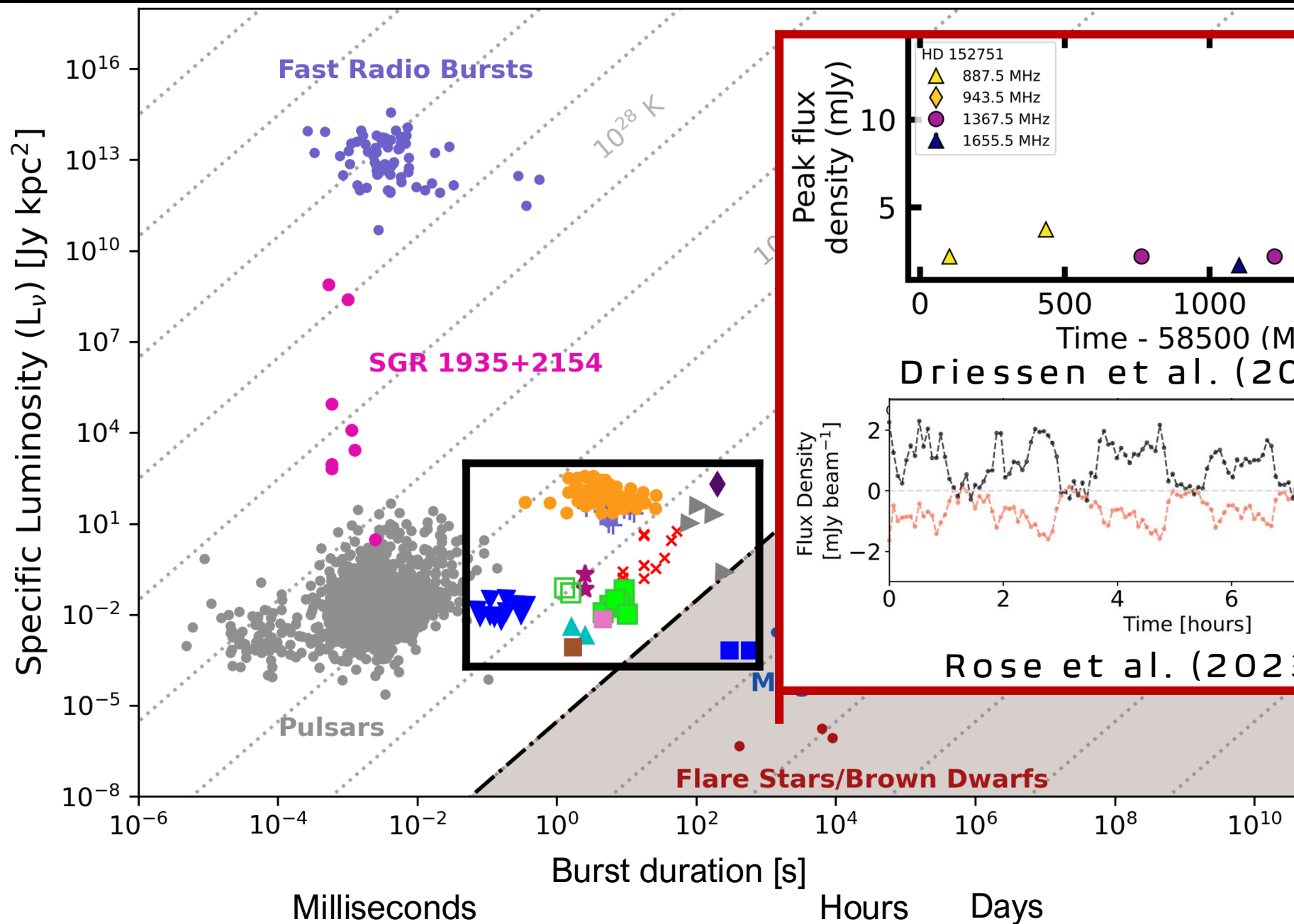
$$16 \pm 6 \text{ pc cm}^{-3}$$

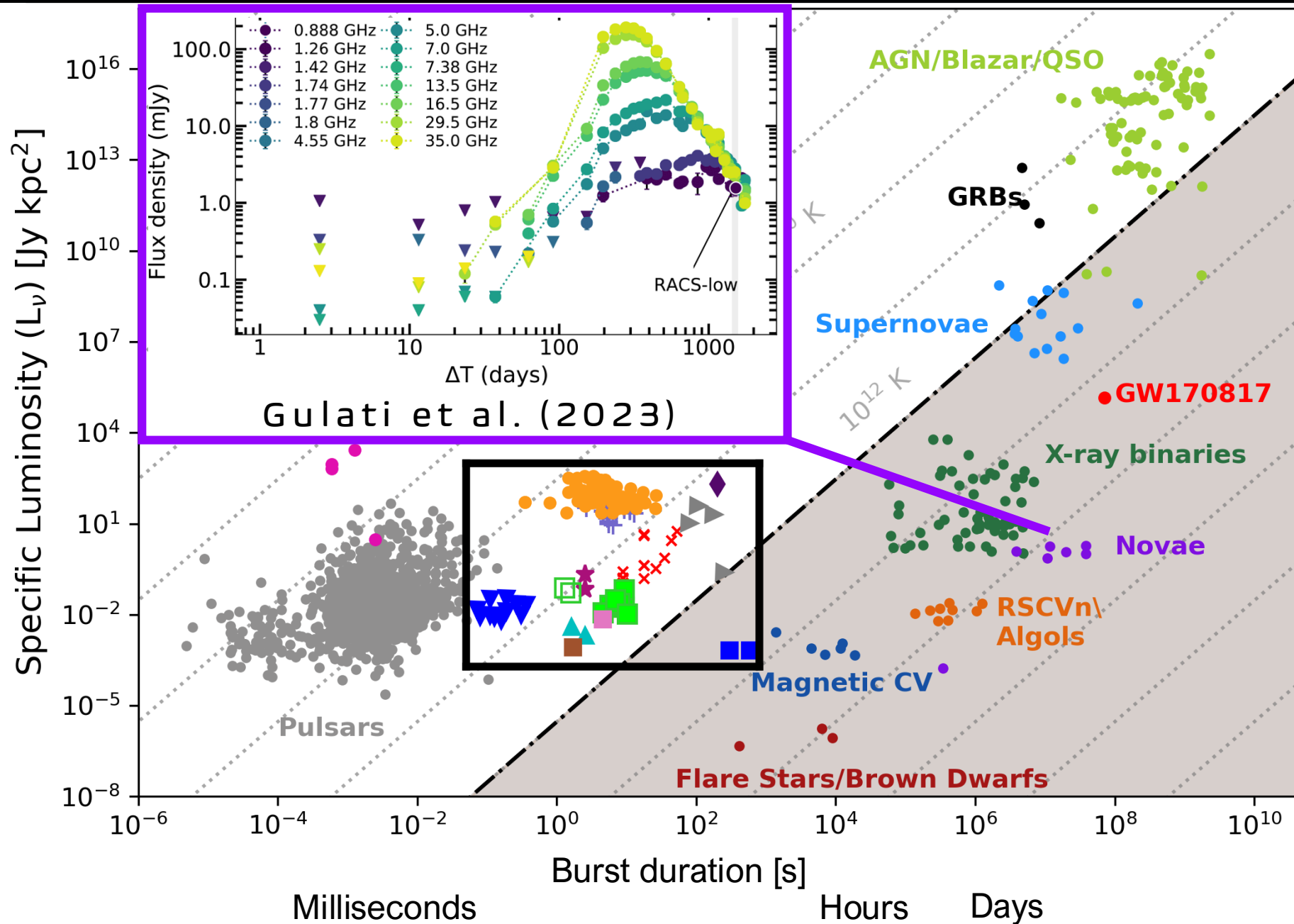


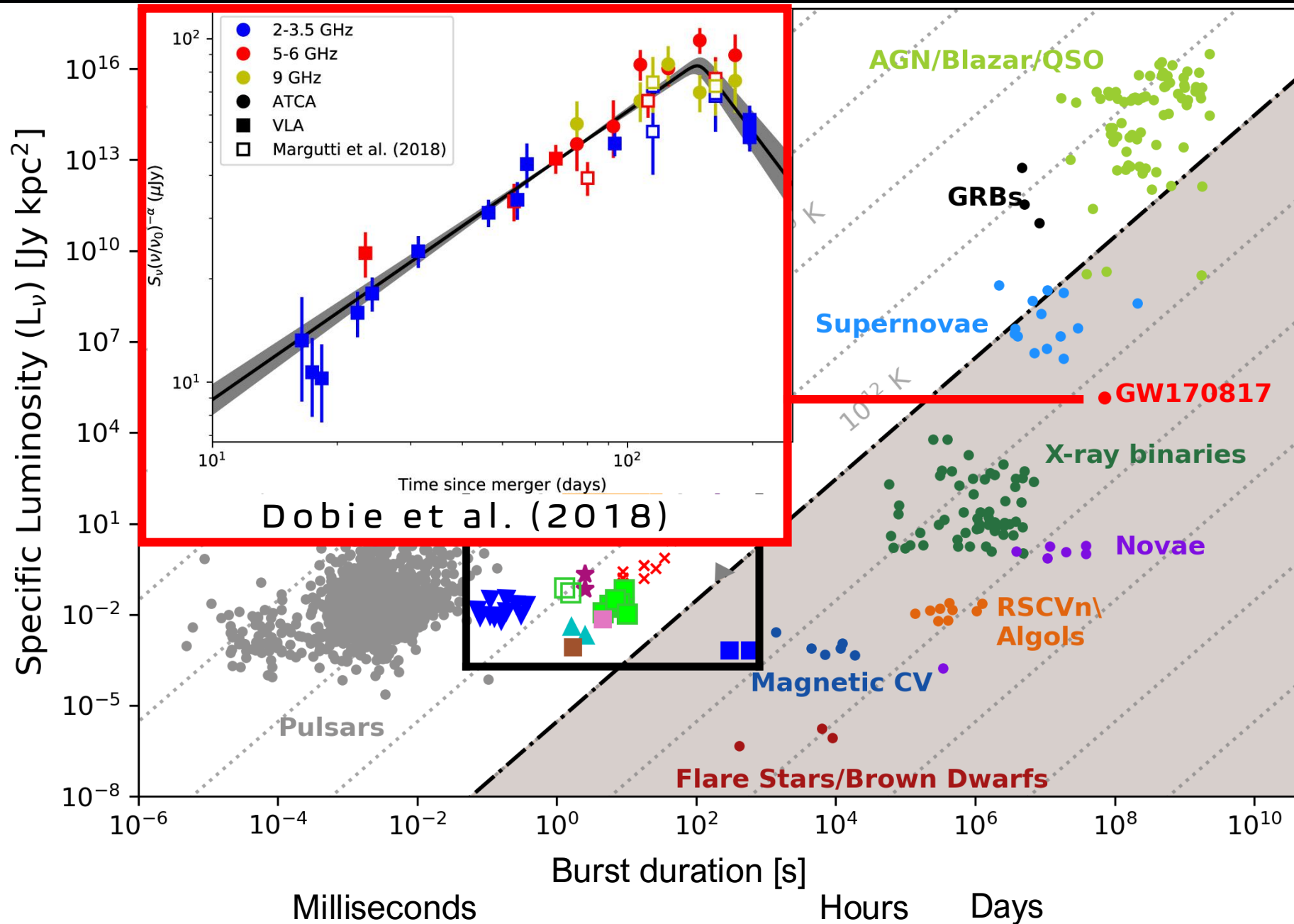
RADIO TRANSIENTS



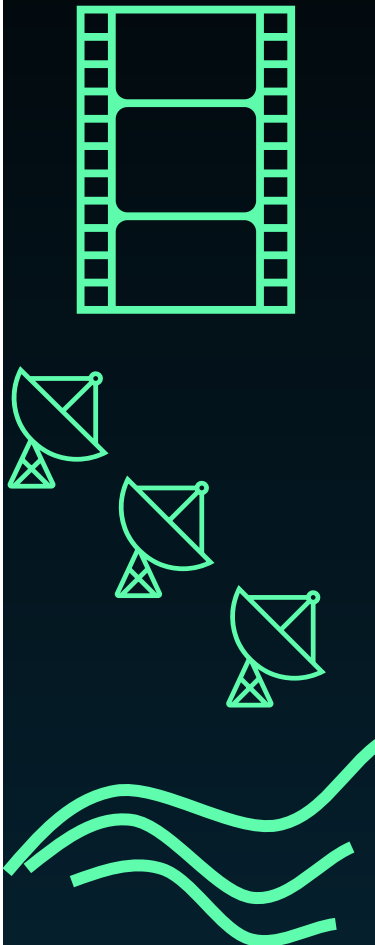
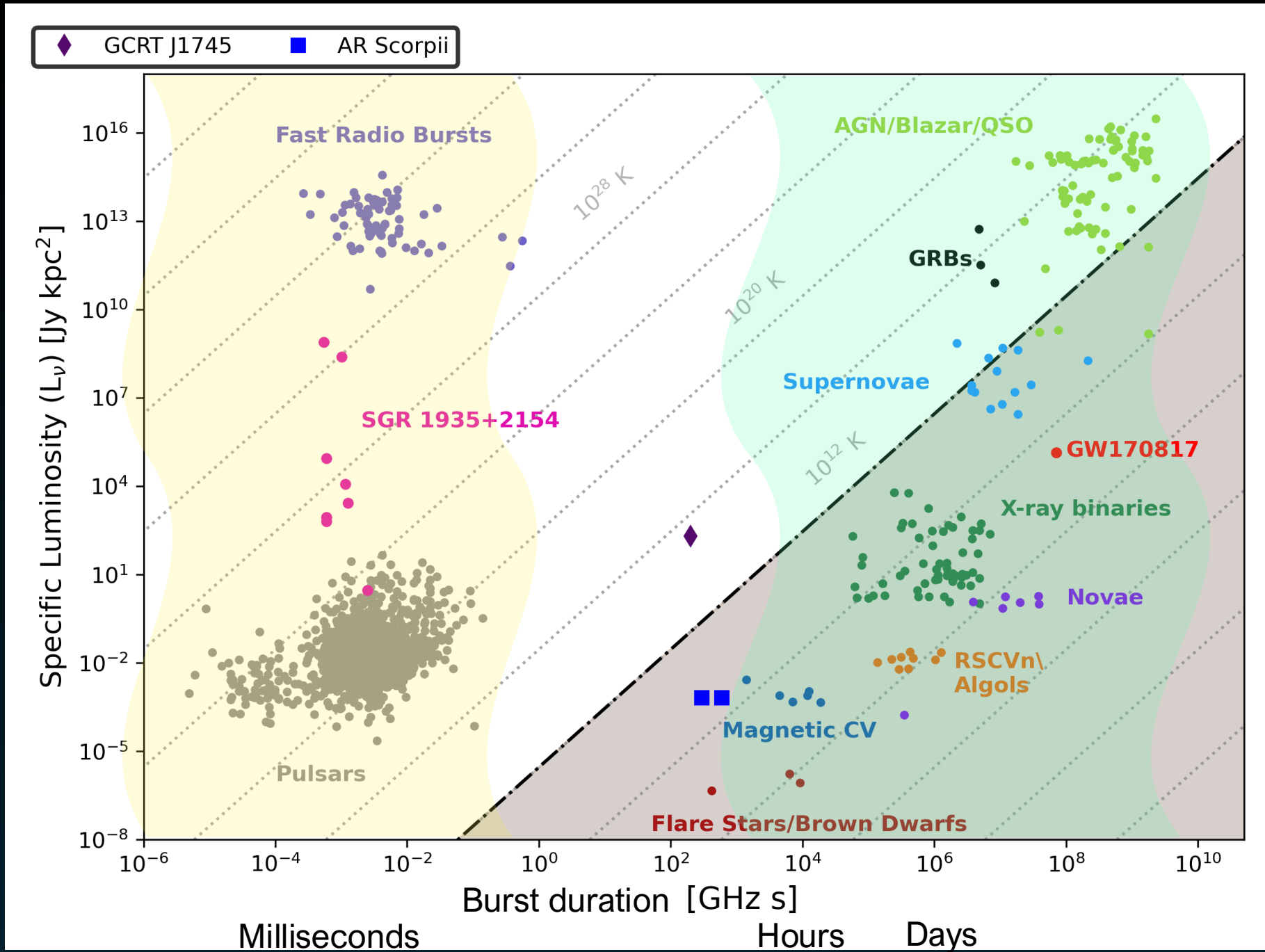


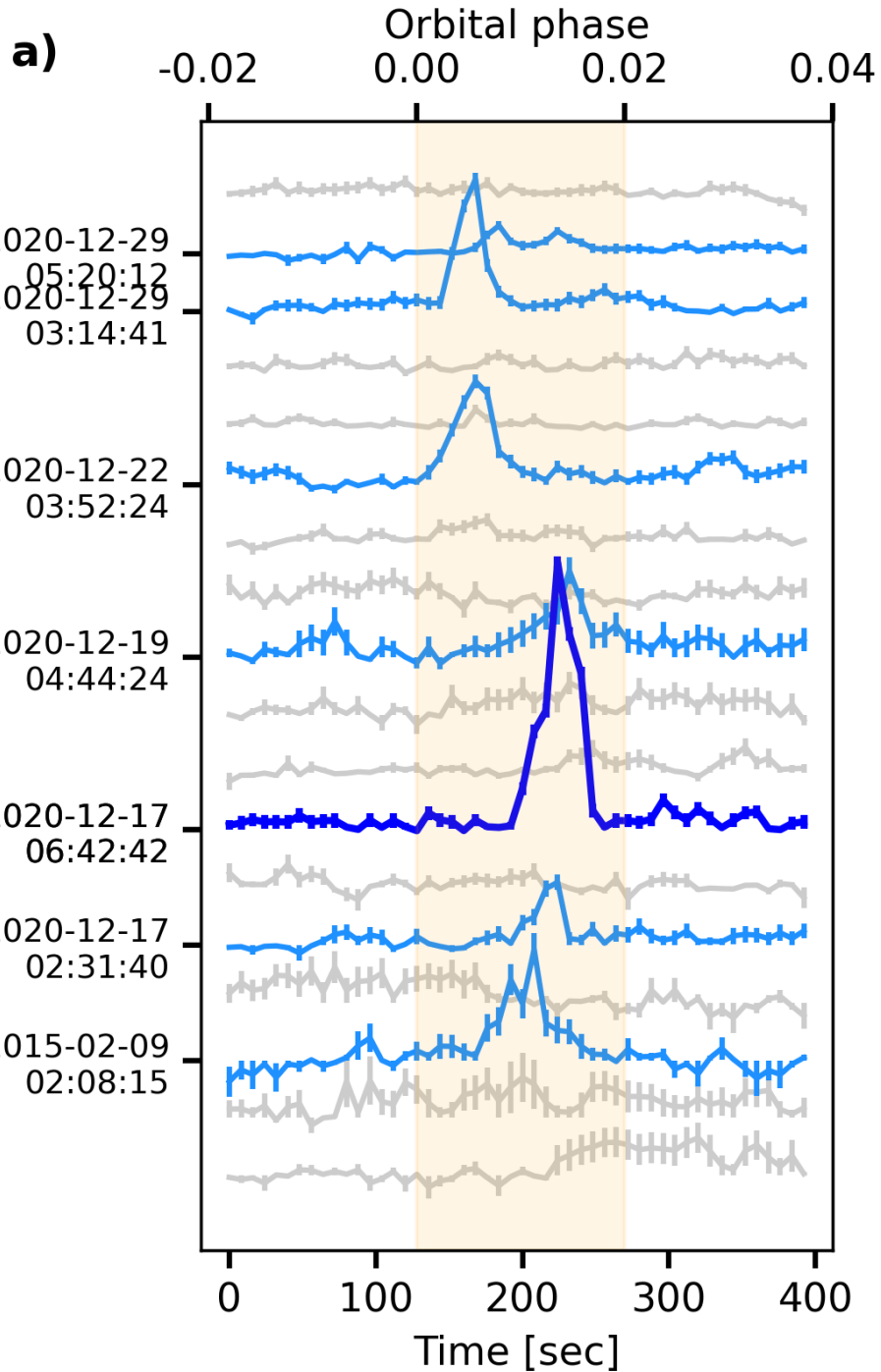






2020





SPORADIC RADIO PULSES FROM A WHITE DWARF BINARY AT THE ORBITAL PERIOD

DE RUITER ET AL (2025)

